

Tutorial

Workflow for the Calculation, Analysis and Mapping of Marine Heatwaves (MHWs) in Xraster

September 2026

Marine Heatwaves (MHWs)

A **marine heatwave (MHW)** is defined as a period during which the local sea temperature exceeds a locally determined threshold, typically the **90th percentile of the local long-term climatology**, for at least **five consecutive days** (Hobday et al., 2016).

The long-term climatology (*T*clim) is the **mean temperature at a given location for each calendar day (day of year)**, calculated over a sufficiently long reference period. A period of **at least 30 years** is generally preferred to obtain a robust long-term climatology.

MHW intensity metrics

Oliver et al. (2018) introduced several measures for characterizing MHW intensity and duration:

- **Duration** – number of days of the MHW (days)
- **Maximum intensity** – maximum temperature anomaly above the climatology (°C)
- **Mean intensity** – mean temperature anomaly above the climatology (°C)
- **Cumulative intensity** – sum of the daily temperature anomalies above the climatology (°C·days)

The standard MHW definition is generally applied **independently at each spatial grid cell**. By combining the affected grid cells, additional measures describing the spatial extent of an MHW can be calculated:

- **Maximum spatial extent** – the largest area affected during an event (km^2)
- **Mean spatial extent** – the mean area affected during an event (km^2)
- **Cumulative area** – the sum of the daily affected areas over the duration of the event (km^2·days)
- **Relative spatial coverage** – percentage of the study area affected by the MHW (%)

MHW categories

An MHW can also be assigned a **category** based on its intensity. The MHW category framework proposed by Hobday et al. (2018) classifies events according to how far the temperature exceeds the **90th percentile threshold**, relative to the local climatology.

The categories are:

- **Moderate** – Category I
- **Strong** – Category II
- **Severe** – Category III
- **Extreme** – Category IV

The category assigned to an MHW event is usually determined by the **highest category reached during any day of the event**. For example, if an event is Moderate for several days but reaches Strong intensity on one day, the event is classified as a **Strong MHW**.

This categorization provides a standardized framework for comparing the severity of MHWs between regions with different background climates. It was introduced by Hobday et al. (2018) and is now widely used in marine heatwave research and operational monitoring.

MHWs as spatio-temporal objects

MHWs can also be treated as **three-dimensional spatio-temporal objects**, with two spatial dimensions and one temporal dimension (longitude × latitude × time). In this approach, an MHW is defined as a **coherent spatio-temporal object formed by the connectivity of anomalously warm grid cells in space and time**.

The object-based approach builds on the local MHW threshold definition. The principal characteristics of an MHW treated as a spatio-temporal object are:

- **duration** – the temporal extent of the object;
- **spatial extent** – the area occupied by the object, which may change over time;
- **intensity** – the magnitude of the temperature anomaly;
- **category** – the MHW intensity category assigned to the object.

Existing tools for MHW analysis

Currently, tools for calculating marine heatwaves are mainly specialized programming packages based on the standard Hobday et al. definition: temperatures exceeding the local 90th-percentile threshold for at least five days.

Commonly used tools include the R package *heatwaveR* , the Python package *marineHeatWaves*, and spatial tracking tools such as *Ocetrac*.

These tools and frameworks can be broadly divided according to how they treat the spatial dimensions of the data.

Pixel-based tools, such as *heatwaveR* and *marineHeatWaves*, calculate MHW statistics independently for each spatial grid cell. For a gridded dataset, the calculation is repeated for every grid cell to identify periods when the local temperature threshold is exceeded.

Object-based tools, such as *Ocetrac*, *TempestExtremes*, and *m_mhw*, consider the spatial field as a whole. They identify connected groups of anomalously warm grid cells and track these groups through space and time. In this way, an MHW can be represented as a coherent three-dimensional structure (longitude × latitude × time) that can change its shape, area, and position, and can potentially split or merge.

Technical and practical limitations

The use of existing MHW tools can involve several technical and practical limitations:

1. **Memory and CPU requirements**
   Pixel-based calculations, such as those implemented in *heatwaveR* or *marineHeatWaves*, require the calculation of local thresholds and MHW events independently for potentially millions of grid cells over decades of daily data. Processing large global datasets can therefore require substantial computational resources and memory.

2. **The 3D big-data problem**
   Object-based approaches such as *Ocetrac* require processing large three-dimensional data cubes (longitude × latitude × time) to identify and track connected spatial structures. Large datasets may therefore require substantial memory and, in some cases, high-performance computing (HPC) or cloud-computing resources.

3. **Programming requirements**
   There is no widely adopted, user-friendly point-and-click **GUI** specifically designed for complete MHW analysis. Many existing workflows require programming skills in R or Python and familiarity with data-processing libraries such as `xarray`, `dask`, and `netCDF4`.

Xraster provides a GUI-based workflow for MHW analysis and mapping directly within QGIS, allowing gridded oceanographic data in NetCDF format to be processed and visualized without requiring the user to develop a complete programming workflow.

The MHW analysis workflow in Xraster consists of **four main stages**:

1. Data preparation

The first stage involves obtaining a **continuous daily sea-temperature dataset**, *T(t, x, y)*, on a regular spatial grid covering the entire study period, including a defined reference period of approximately **30 years**.

The data are prepared in **NetCDF format**, which is a standard format for multidimensional oceanographic data, such as datasets available from the Copernicus Marine Data Store.

2. Calculation of MHW variables and masks

The second stage involves calculating the following variables:

- **temperature anomaly**;
- **MHW category**;
- **MHW object ID**;
- **MHW object mask**.

The MHW object mask is defined according to the selected criteria for **object duration and spatial extent (area)**.

An **AOI (Area of Interest) mask** is also created to define the spatial area included in the analysis.

3. Creation of the MHW dataset and object descriptions

In the third stage, a new dataset is created containing the original **sea temperature** and the calculated MHW variables.

In addition, a series of **text files** is generated containing information describing individual MHW objects, including their temporal and spatial characteristics.

4. Visualization and analysis

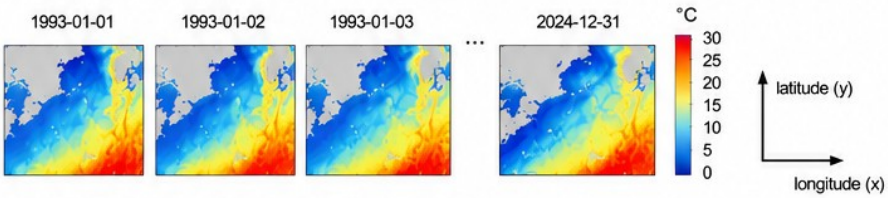
The final stage consists of the **visualization and analysis of MHWs in Xraster**, using the created dataset and the accompanying text files.

The implementation of this workflow in Xraster is presented in the figure below.

1 Daily $T(t, y, x)$ NetCDF

- Daily sea temperature (°C)
- 3D array: time × latitude × longitude
- Example: 1993-01-01 ... 2024-12-31

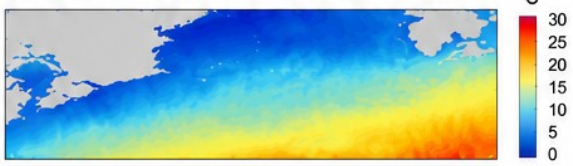
Example: daily sea temperature



2 Climatology (e.g. 1993–2022)

- Daily climatology (mean seasonal cycle)
- Computed for each grid cell

Example: climatological mean (day of year = 200)



For each grid cell and day of year (1–366): mean over 1993–2022

3 90th percentile threshold

- For each day of year and grid cell, compute the 90th percentile
- 1993–2022

Example: 90th percentile threshold (day of year = 200)

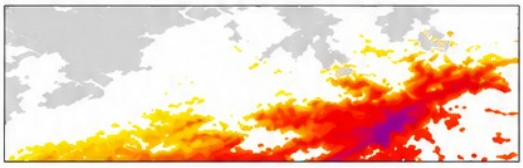


For each grid cell and day of year (1–366): 90th percentile over 1993–2022

4 Categories MHW mask

- Where daily T > threshold, assign intensity categories

Example: MHW categories on 2024-08-15

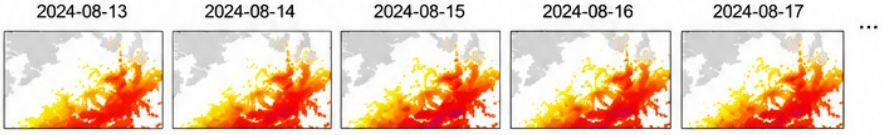


0	No MHW
1	Category I (moderate)
2	Category II (strong)
3	Category III (severe)
4	Category IV (extreme)

5 Daily category maps (2D)

- Create daily 2D maps of MHW categories

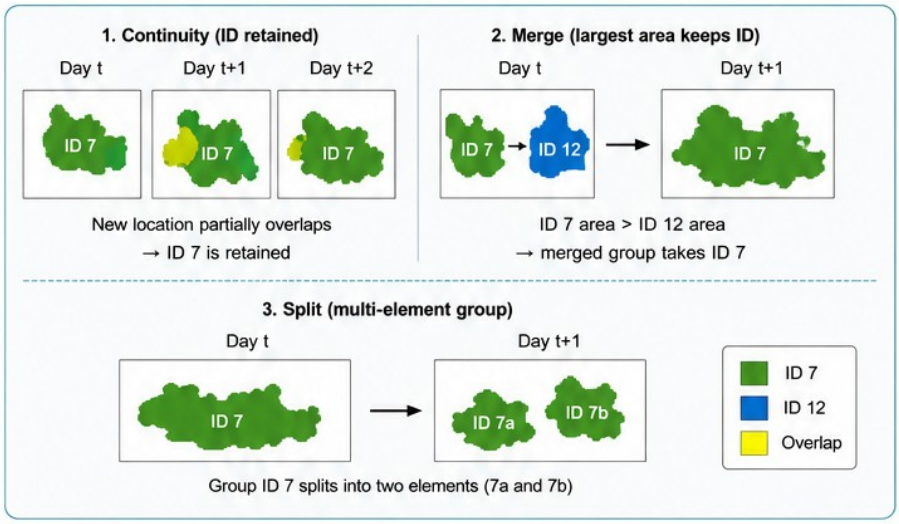
Example: sequence of daily category maps



6–7 ST Region Group (2D)

Spatio-temporal region groups

- Spatial-temporal region groups are created from binary data (0,1) separately for each category
- Groups maintain continuity in both space and time
- Each group is assigned a unique ID
- ID is retained if the new location partially overlaps with the previous position
- When connected groups with different IDs merge, the resulting group takes the ID of the group that contributed the largest area
- A group can split, forming a multi-element group



7 Output: MHW object table

- Summary of all detected MHW objects

ID	Category	Start date	End date	Duration (days)	Maximum area (km²)	Maximum area date	Maximum intensity (°C above threshold)	Mean intensity (°C above threshold)
7	III (severe)	2024-08-10	2024-08-23	14	85,420	2024-08-15	3.21	1.42
12	II (strong)	2024-08-11	2024-08-16	6	31,680	2024-08-14	2.15	1.08
7a	III (severe)	2024-08-24	2024-09-02	10	46,220	2024-08-27	2.78	1.35
7b	III (severe)	2024-08-24	2024-08-31	8	28,910	2024-08-25	2.34	1.16
15	IV (extreme)	2024-08-18	2024-08-26	9	52,110	2024-08-20	3.95	1.76
...	...	...	...	...	...	...	...	...

Cumulative MHW Category Approach

In the MHW analysis, a **cumulative approach to MHW intensity categories** was applied.

MHW parameters were calculated using cumulative intensity categories: Category 1 (Moderate) includes MHWs with intensity > 0; Category 2 (Strong) includes MHWs with intensity > 1; Category 3 (Severe) includes MHWs with intensity > 2; and Category 4 (Extreme) includes MHWs with intensity > 3.

Thus, the categories are **cumulative rather than mutually exclusive**. A Category 3 MHW is therefore also included in Categories 1 and 2, while a Category 4 MHW is included in Categories 1, 2, and 3.

The workflow starts by loading a **daily temperature NetCDF dataset** with two spatial dimensions and one temporal dimension (2D space × time) **(1)**.

The dataset may contain **sea surface temperature (SST)**, **bottom temperature**, or **mean temperature for a specified water layer**. The dataset may also include a **mask defining the area of interest**.

Next, an **MHW category subset** is created, containing values from **0 to 4**, where 0 represents no MHW and values 1–4 represent the corresponding MHW intensity categories.



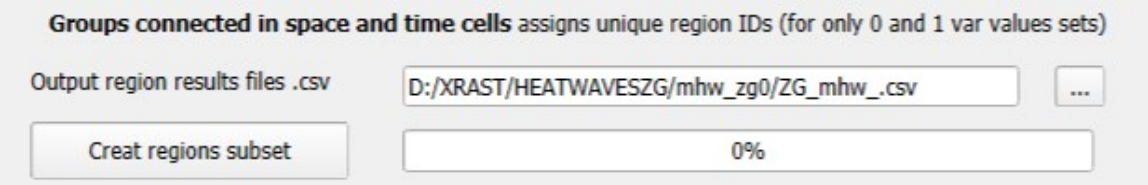
The climatology is calculated from the temperature dataset (2) for the selected reference period (e.g., 1993–2024), together with the 90th-percentile threshold (3).

These two variables are then used to detect MHWs according to the **standard MHW definition (4)**. The resulting subset contains values **1, 2, 3, and 4**, corresponding to the respective MHW intensity categories. Areas and periods where no MHW is detected are assigned a value of **0**.

The resulting MHW category subset can be further analyzed and saved as a **NetCDF dataset**.

Creation of MHW objects

The next step is to identify and create **individual MHW objects (6–7)**. Each object represents a spatially and temporally coherent MHW and is assigned a unique **MHW object ID**.



MHW Object Identification

The connected cells are then **grouped according to their spatial and temporal connectivity**, and each group is assigned a unique **integer ID**. Each ID represents an individual MHW object.

The results are stored as a new subset and an accompanying **text file describing each MHW object**. The resulting subset can be further analyzed and saved as a NetCDF dataset.

Final MHW Dataset

After the required variables have been calculated, a new dataset is created containing the following variables:

- **temperature**
- **day-of-year anomaly**
- **marine heatwave category**
- **MHW object ID**
- **MHW object mask**

An accompanying **text file** contains a description of the identified MHW objects and their characteristics.

The **marine heatwave category** and **MHW object ID** variables are derived solely from the temperature data and the corresponding climatology and threshold. At this stage, every detected MHW object is identified regardless of its duration or spatial extent.

The classification of an object as a **significant MHW** is determined by additional criteria, primarily its **duration and spatial extent (area)**. These criteria are applied using the **MHW object mask**, which contains binary values (**0/1**):

- **0** – the MHW object does not meet the specified duration and/or spatial-extent criteria;
- **1** – the MHW object meets the specified criteria and is included in the analysis.

Extended MHW Analysis Using Python Scripts

An extended analysis of MHWs is possible through the creation of a series of **data products** derived from the MHW dataset and object descriptions.

These products include:

1. **A table containing the characteristics of individual marine heatwave events (MHW objects).**

	Category	Peak Date	Duration (days) and Time	I_{max}	I_{mean}	T_{max}	T_{mean}	CI	D moderate [days] / max area [km^2]	D strong [days] / max area [km^2]	D severe [days] / max area [km^2]	D extreme [days]/ max area [km^2]	Season
1	Category III (Seviere)	1993-5-21	36 / 1993-4-25 - 5-30	6.49	2.92	19.62	11.64	105	121/ 4661	12 / 579	3 / 43	-	Spring

MHW Object Characteristics Table

The table contains the following columns:

- **Category** – MHW category
- **Peak date** – date on which the MHW reached its maximum spatial extent
- **Duration** – duration of the MHW, including the start and end dates (number of days)
- **Imax** – maximum temperature anomaly above the climatology (°C)
- **Imean** – mean temperature anomaly above the climatology (°C)
- **CI** – cumulative intensity, calculated as the sum of daily temperature anomalies above the climatology (°C·days)
- **Dmoderate** – number of days and maximum spatial extent of the Moderate category
- **Dstrong** – number of days and maximum spatial extent of the Strong category
- **Dsevere** – number of days and maximum spatial extent of the Severe category
- **Dextreme** – number of days and maximum spatial extent of the Extreme category

Description of MHW Objects

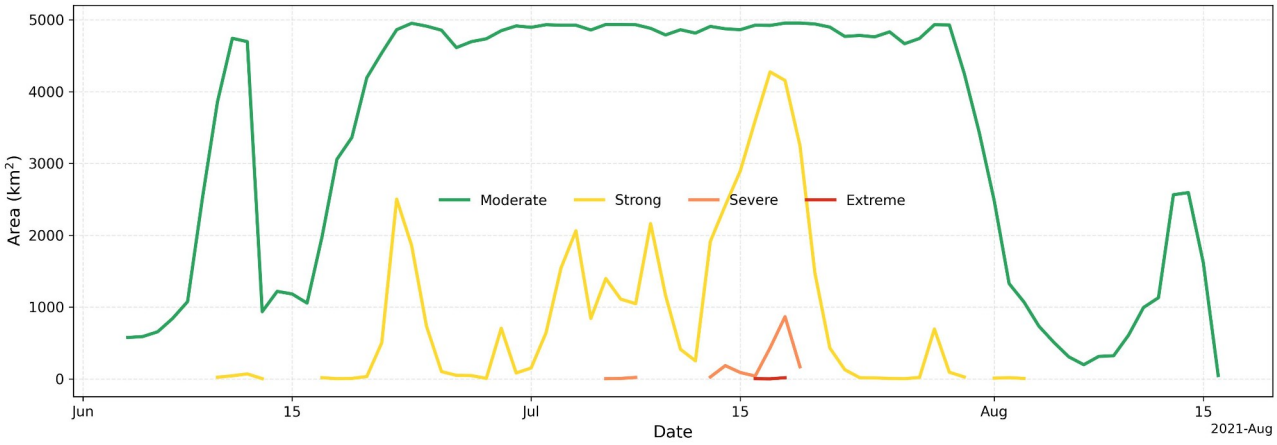
Marine heatwaves are treated as **spatio-temporal objects**. Their characteristics can be described using intensity statistics, such as **cumulative, mean, and maximum intensity**, together with

temporal parameters such as **initial detection date, duration, timing of peak intensity, and timing of maximum spatial extent**.

Together, these parameters provide a comprehensive quantitative description of individual MHW events within a given study area. They allow individual heatwave events to be identified and selected for **further, more detailed analysis**.

2. Time Series of Changes in the Spatial Extent of Individual MHW Categories

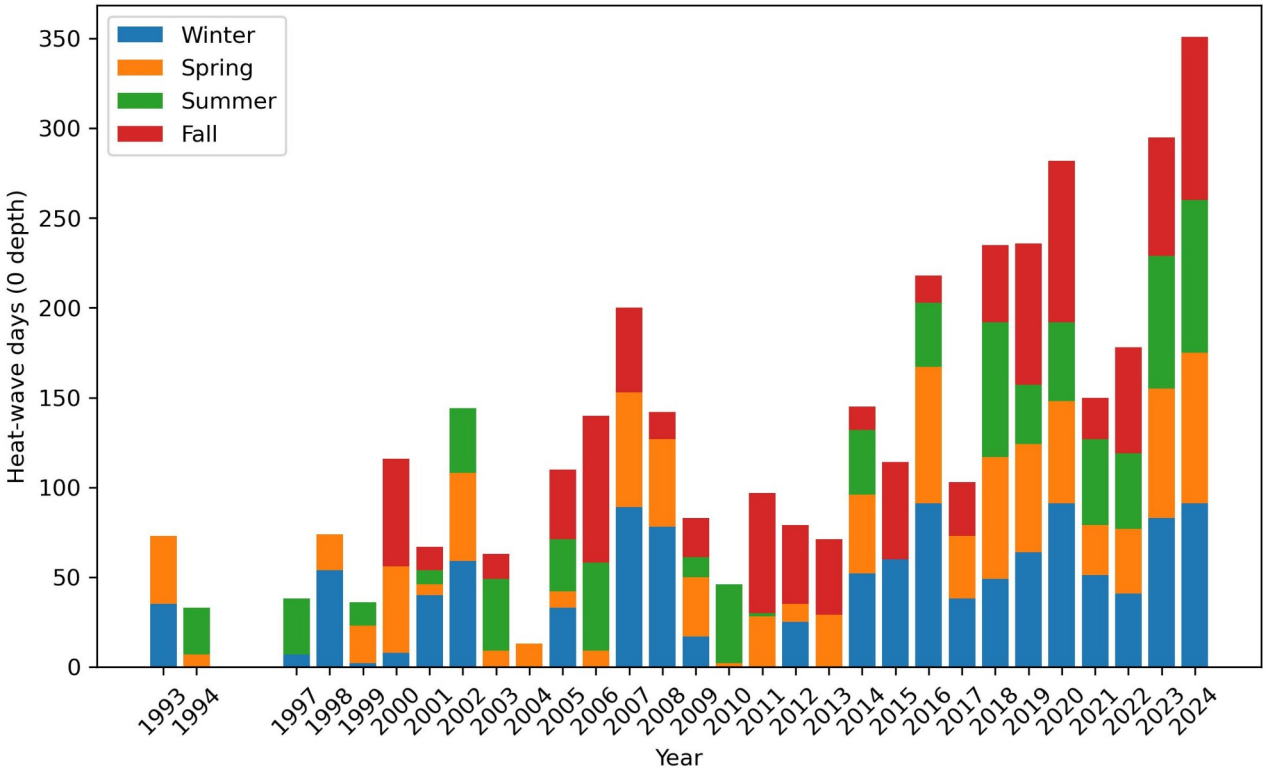
A **time series showing changes in the spatial extent (area) of individual MHW categories** during the analyzed event.



The graph allows analysis of **changes in the spatial extent of a given MHW over time**. Including areas belonging to higher MHW categories makes it possible to visualize **changes in the internal structure and intensity of the MHW throughout the event**.

When presented together with a map showing the **maximum MHW category**, the time series provides a **spatio-temporal representation of the MHW**, showing both how its spatial extent changes over time and how its intensity structure develops during the event.

3. Number of MHW Days in Each Season for Individual Years



This graph enables analysis of **temporal variability in MHW occurrence by season**. It also provides a clear way to identify and assess **temporal trends in MHW occurrence**.

MHW Analysis Step by Step

We will conduct an example MHW analysis for a part of the Mediterranean Sea using sea surface temperature data obtained from the Copernicus Marine Data Store.

The data were downloaded using the Python Application Programming Interface (API) with a dedicated script.

```
# Storing directory
DataPath = r'D:\XRAST\wersja5\MSHW'

# File name
FileName = 'MS_30years.nc'

copernicusmarine.subset(
  dataset_id="cmems_SST_MED_SST_L4_REP_OBSERVATIONS_010_021",
  variables=["analysed_sst"],
  minimum_longitude=16.04494268732859,
  maximum_longitude=21.29678673772561,
  minimum_latitude=36.867424180605745,
  maximum_latitude=40.87541042959295,
  start_datetime="1996-01-01T00:00:00",
  end_datetime="2025-12-31T00:00:00",
  output_filename = FileName,
  output_directory = DataPath
)
```

The **MS_30years.nc (180 MB) dataset can be downloaded using link**:

https://drive.google.com/file/d/1wJIQJp8bde9Ff8euJLRr5pT4L8CK3Y9Y/view?usp=sharing

Additionally, two Python scripts are used:

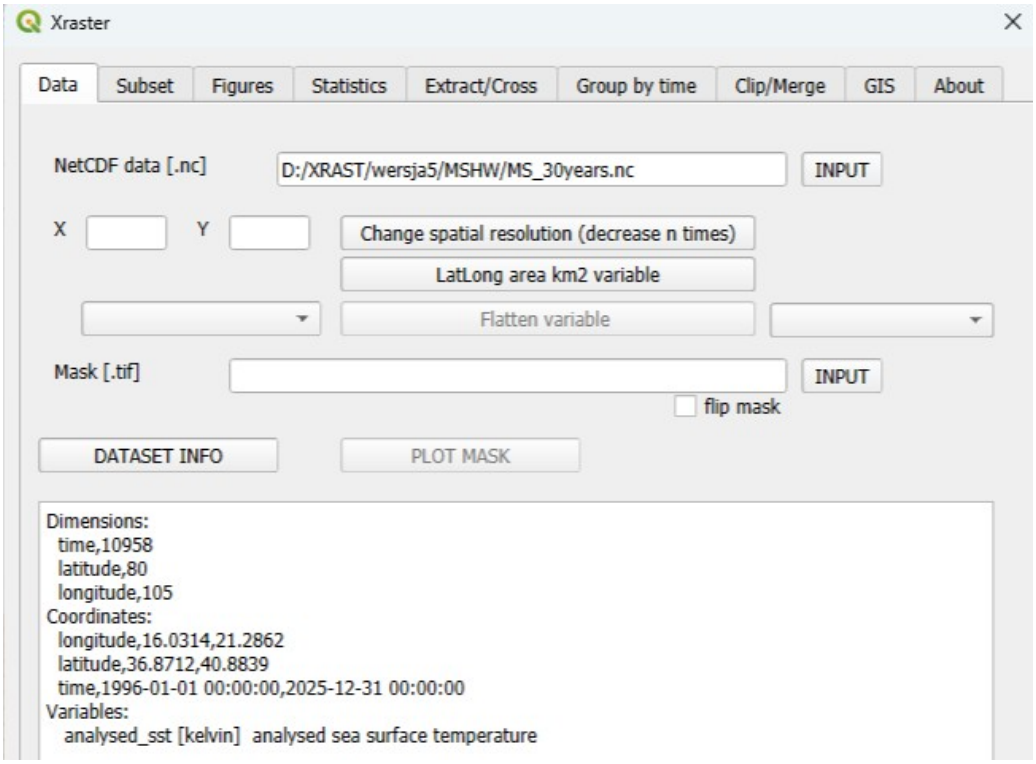
- **MHW_ext_Script_1.py**
- **MHW_ext_Script_10.py**

The **Jupyter Notebook *MHW_extension.ipynb*** is also used.

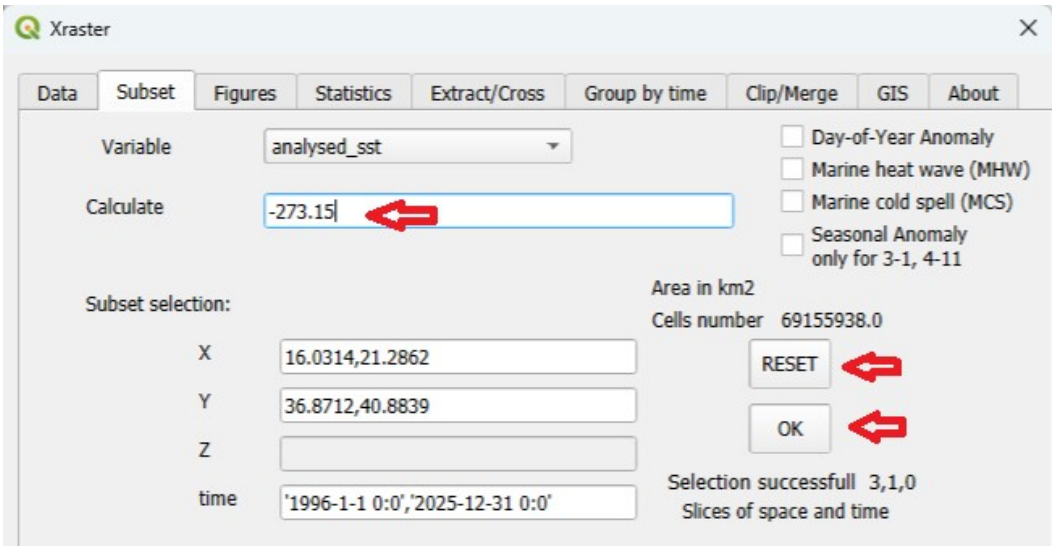
All these files are available for download as **MHW_python.zip from**

https://github.com/urbanskigis/Xraster

We begin by **importing the data into Xraster** and becoming familiar with the dataset through **exploratory data analysis (EDA)**.

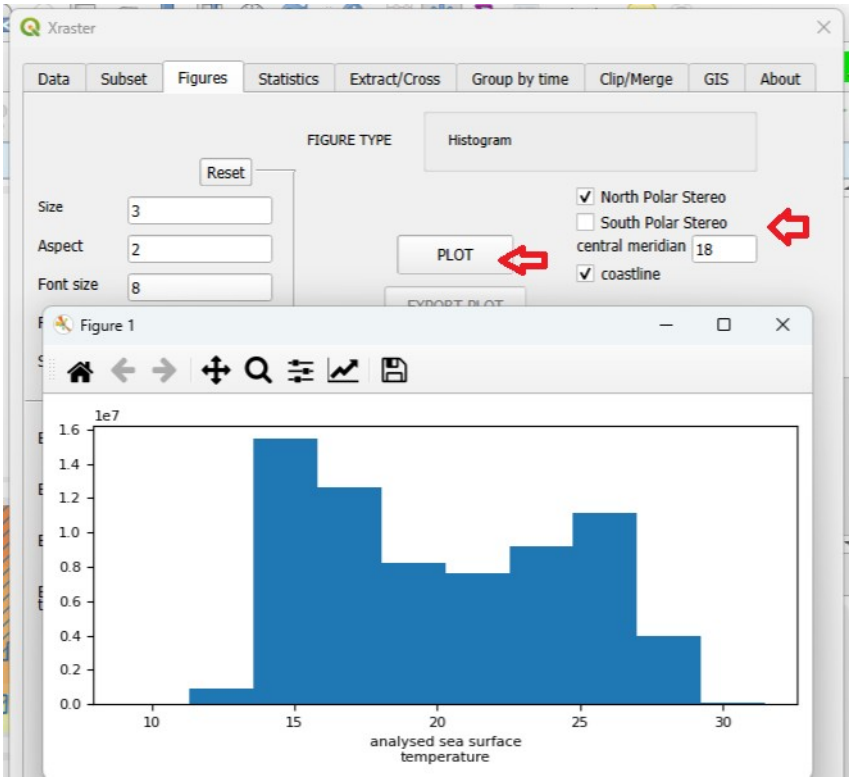


We create a **SUBSET for temperature** and convert the temperature units to **degrees Celsius**.

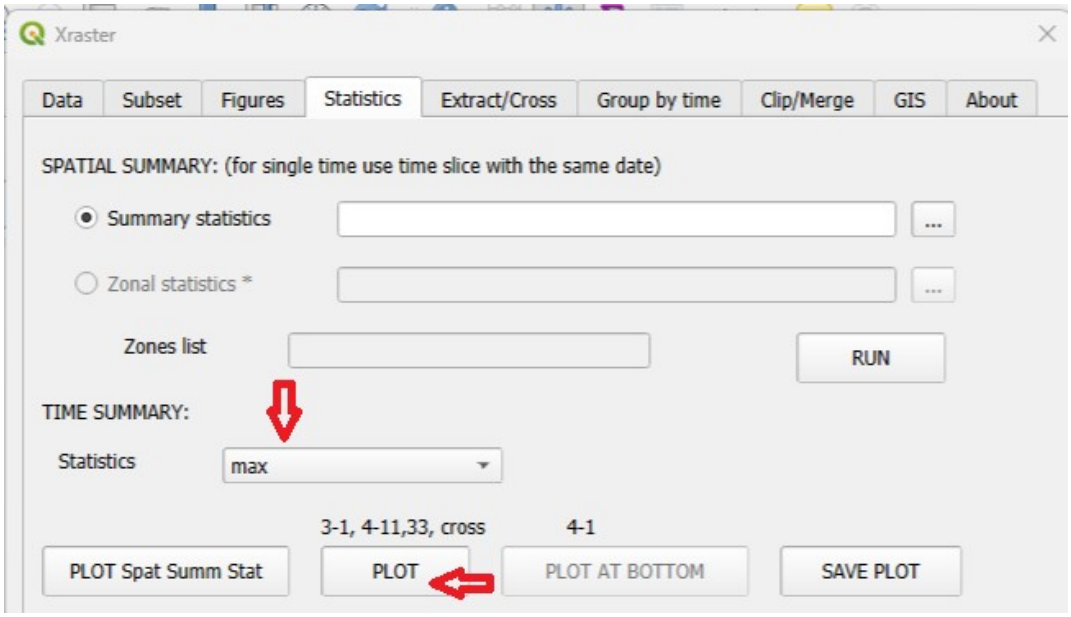


We can now create a **temperature histogram** to examine the temperature range.

Additionally, we specify the **map projection parameters** and the settings for **drawing the coastline**.

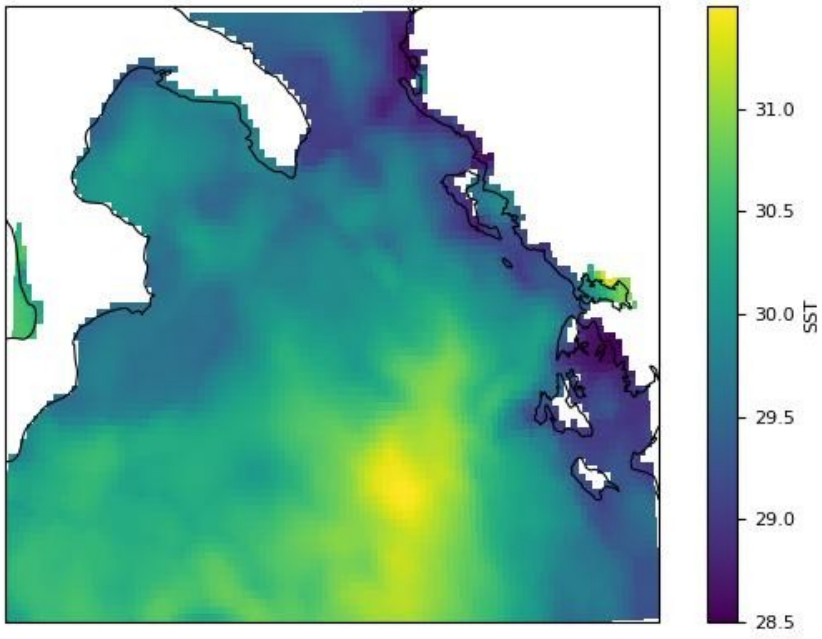


We will create a **map of the maximum temperature values**.

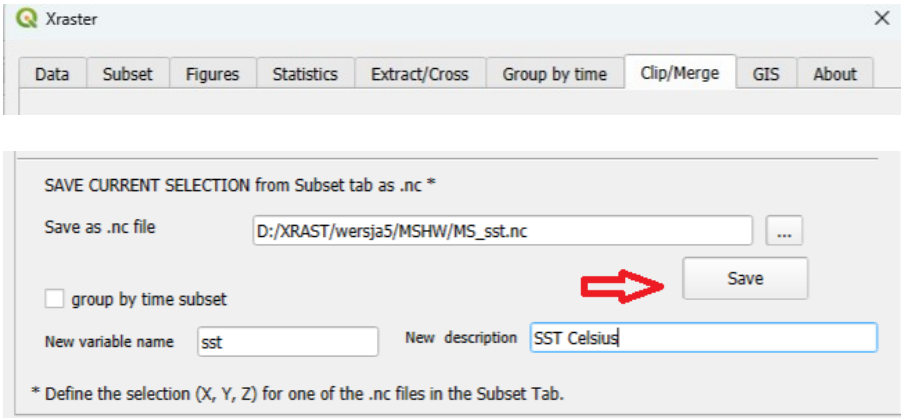


Creating the map we also visualize the **analysis area**, covering the northern part of the Ionian Sea and the coasts of Italy, Albania, and Greece, using the **maximum temperature values**.

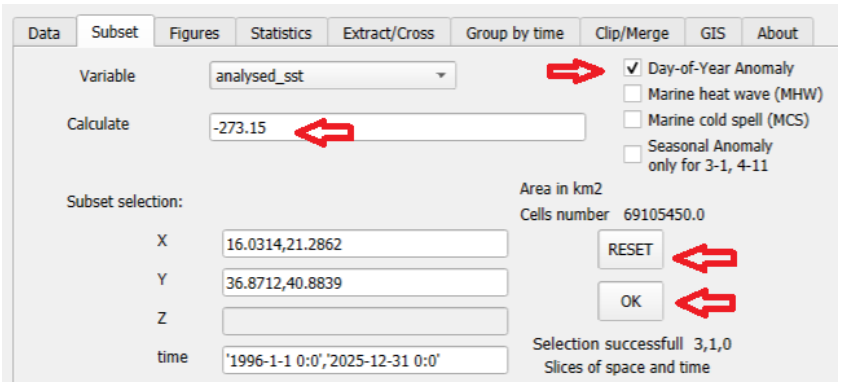
The map was created using the **save (floppy disk) icon** on the figure.



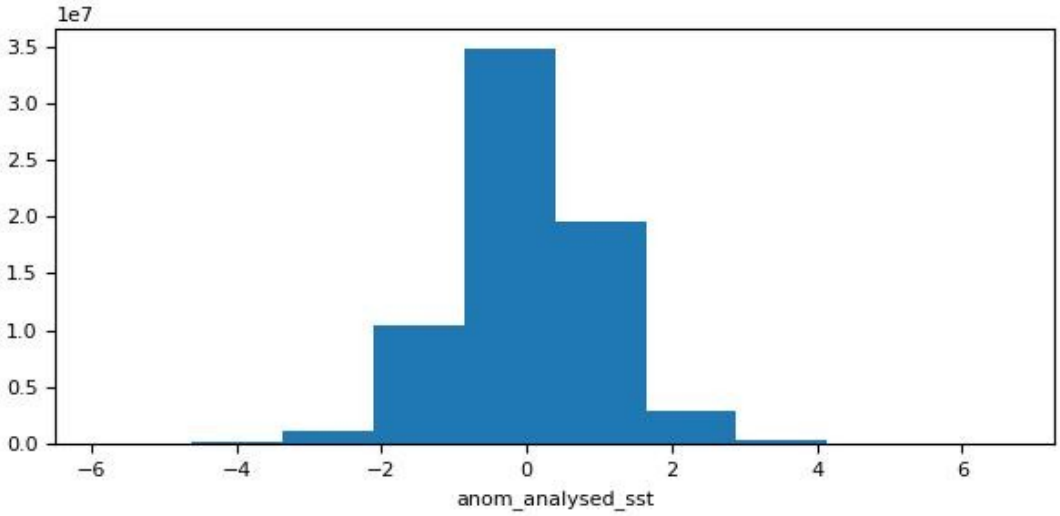
The created **SUBSET** is saved as a new **NetCDF file**.

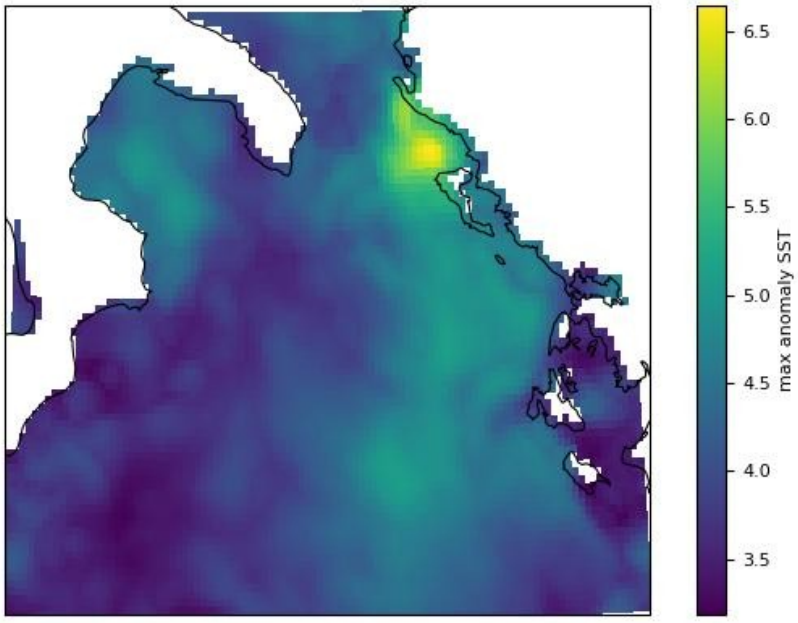


We create a **SUBSET of the Day-of-Year Anomaly**.

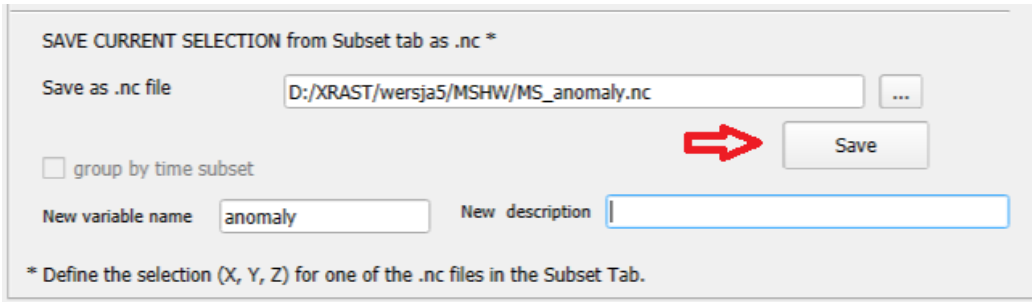


As before, we create a **histogram** to examine the range of anomaly values and a **map of the maximum anomalies**.

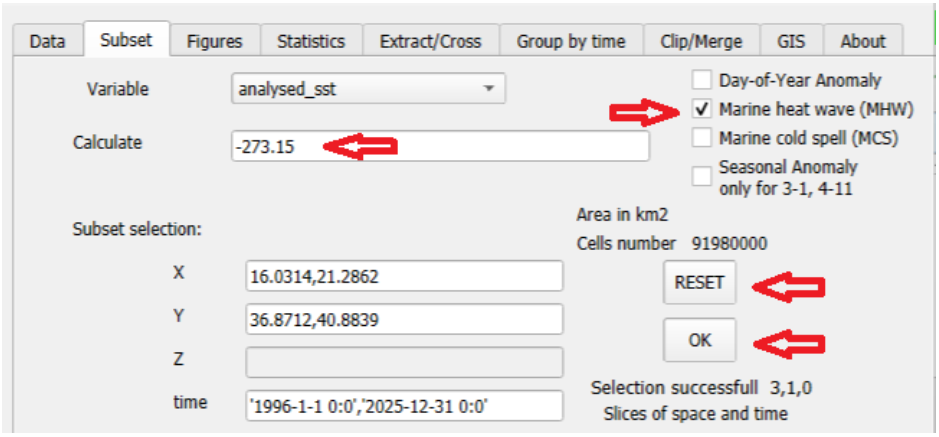




As before,
we save the created **SUBSET** as a new **NetCDF file**.



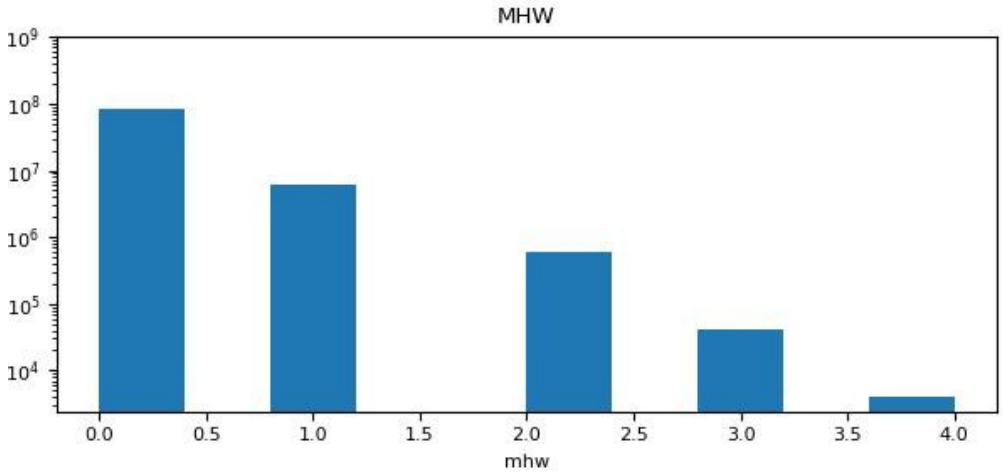
We create a **SUBSET Marine Heat Wave**. It contains four MHW categories (1, 2, 3, and 4), while 0 indicates no MHW. The categories refer to temperature anomalies, in accordance with the definition of Hobday et al. (2016), but do not include the requirement that the event lasts for at least 5 days.



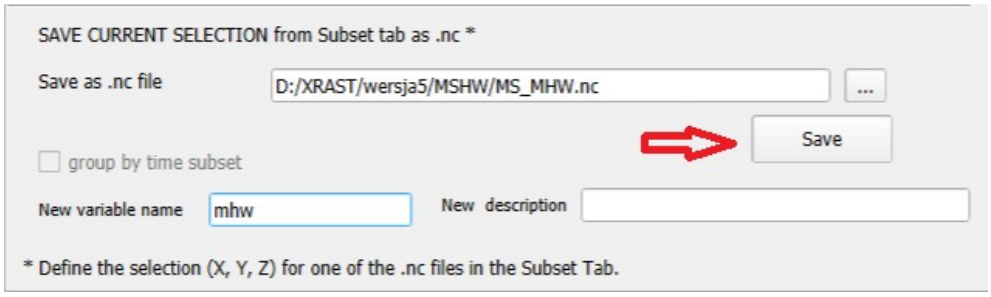
Note! This operation may take several minutes to complete.

We will create a histogram to check the resulting values (a logarithmic scale is used on the Y-axis). Click the second-to-last icon 

and select a logarithmic scale for the X-axis. You should also increase **Top** to **1e9**.

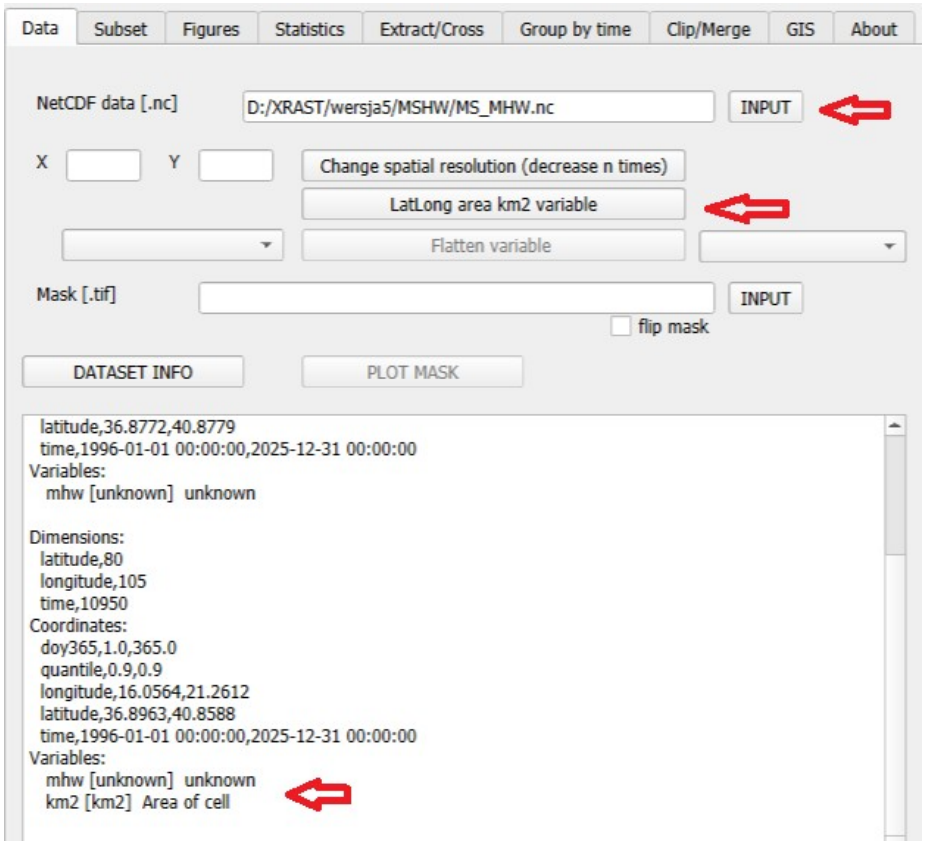


We save the completed **SUBSET** as an `.nc` file.



We will calculate the area of the map cells in km².

We open the **MS_MHW.nc** file and click the **LatLong area km2 variable** button, creating an additional variable containing the cell area in km².



Since we will need the area of a single cell in km², we will use these data to calculate it.

We create a **SUBSET** for a single day by entering the following expression in the **Calculate** field:

```
0 + ds2.km2
```

Our raster algebra expression will therefore take the form:

```
mhw * 0 + km2
```

(where **mhw** and **km2** are variables).

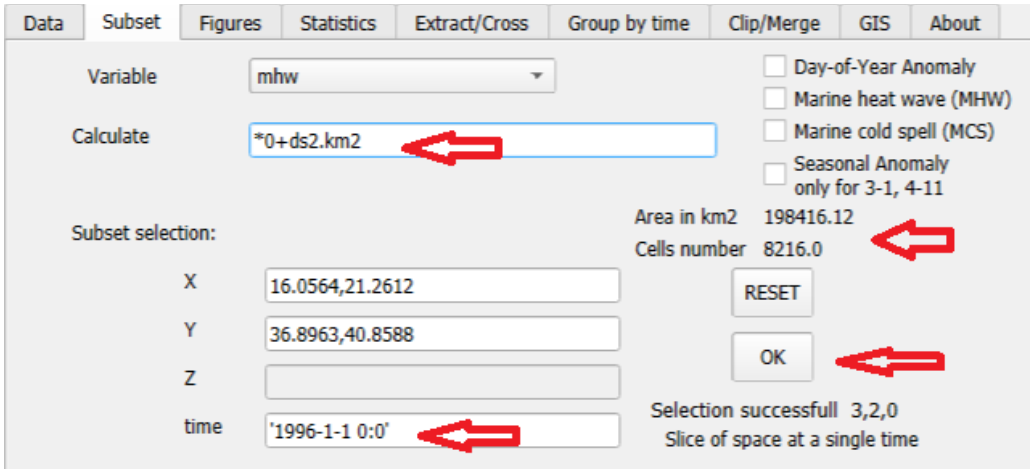
Thus, the area of each cell will be assigned to the corresponding cell in square kilometres.

The panel will display the number of cells and their areas.

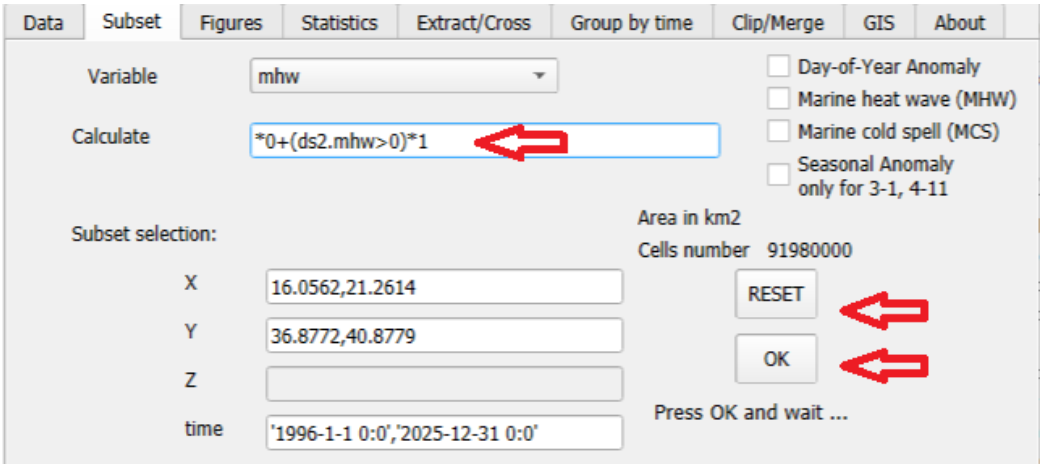
Finally:

198416.12 / 8216 = **24.15**

(the mean area of a map cell in km²).



We will now create a **SUBSET** for the MHW objects ID. We will convert the values 1, 2, 3, and 4 to 1 using a raster algebra expression.



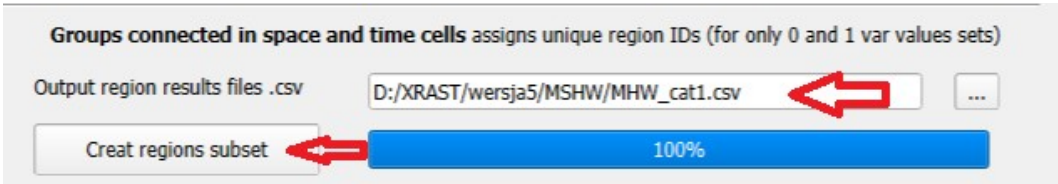
The expression is:

```
mhw * 0 + (mhw > 0) * 1
```

(In the **Calculate** field, we enter the part of the expression following the variable name.)

If **mhw** is greater than zero (logical value = 1), the result is 1; otherwise, the result is 0.

We open the **GIS** tab, enter the name of the text file containing the results of the operation, and click the **Create regions subset** button.



The progress of the calculations can be monitored on the progress bar, which shows the percentage completed.

A new **SUBSET** is created containing the object IDs (**region ID**) and two text files:

MHW_cat1.csv and
MHW_cat1_sum.csv.

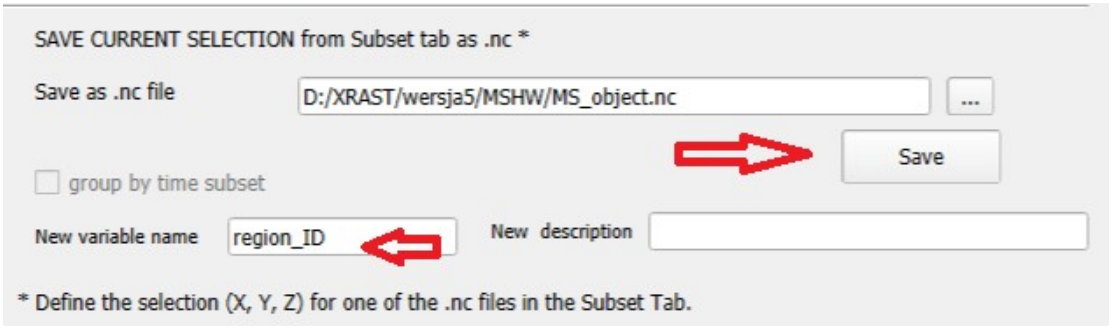
The **MHW_cat1_sum.csv** file contains information about the identified objects, sorted by their maximum area. It has the following form:

id,max_number_of_cells,date_start,date_end,date_of_max
32835,6268,2018-04-08,2018-06-21,2018-05-30
6640,6268,2001-02-26,2001-04-13,2001-03-28
47244,6268,2023-07-09,2023-08-07,2023-07-21
50585,6268,2024-03-08,2024-04-25,2024-04-08
51900,6268,2024-07-08,2025-05-26,2024-07-17
16801,6256,2009-05-19,2009-06-03,2009-05-25
43211,6247,2022-05-14,2022-06-23,2022-06-04
22318,6247,2013-04-24,2013-05-14,2013-05-03
43657,6240,2022-06-19,2022-07-12,2022-07-04

.....

This file contains more than 20,000 objects, but because they are sorted by maximum area (**max_number_of_cells**), we can easily identify the most significant objects in terms of area. We also have information about when each object started and ended, as well as the date on which it reached its maximum area.

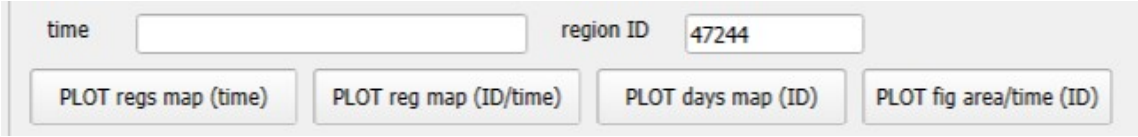
First, however, we will save it as a fourth file containing the **region_ID** variable.



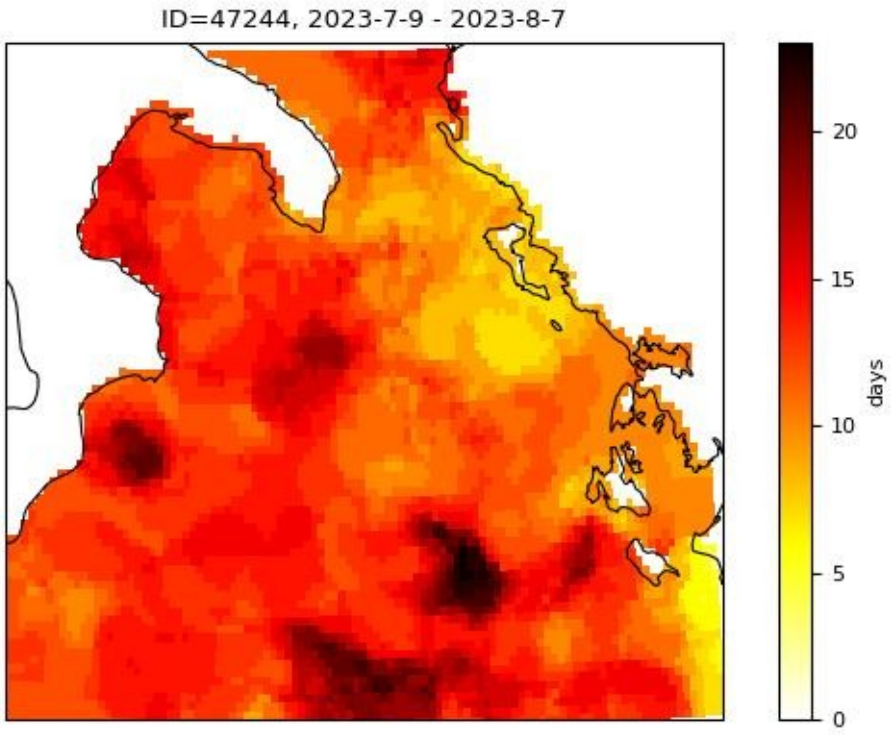
Since the region ID subset remains in memory after completing the procedure, we can perform a series of analyses using the tools available in this tab.

We will analyse the fourth object, with **ID = 47244**.

The tool provides four buttons that enable the visualization of objects according to time and their ID.

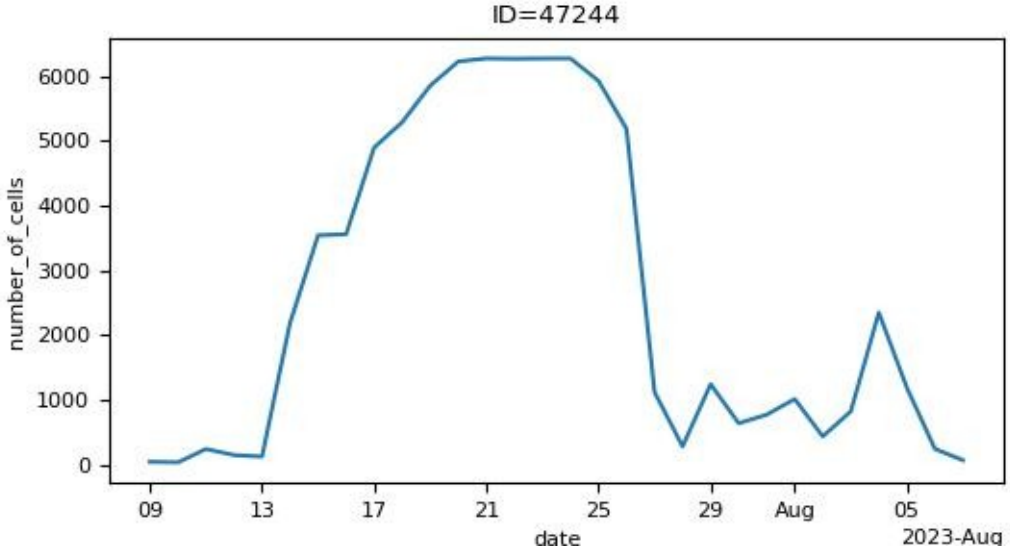


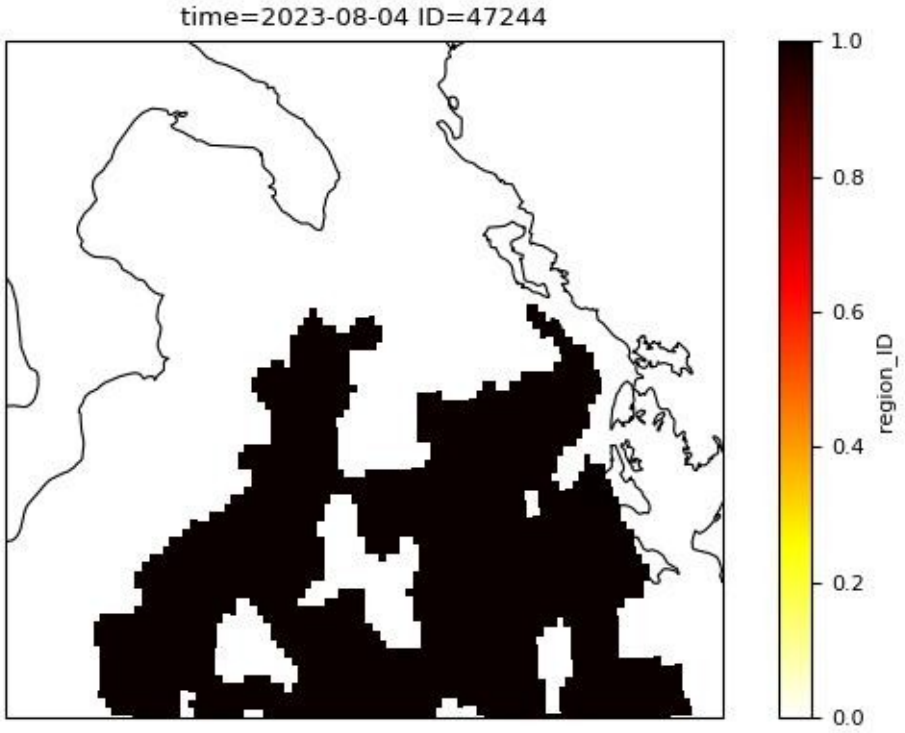
Entering the **region ID** and clicking the **PLOT days map** button produces a map showing the number of days for which the given MHW (ID = 47244) existed in each area.



Entering the **region ID** and clicking the **PLOT fig area/time** button produces a plot showing changes in the area of the given MHW (ID = 47244) over time.

Entering the ID and date (e.g. 2023-08-04) allows us to obtain the location of the MHW object on a given day by clicking the PLOT reg map (ID/time) button.





We repeat the **region groups** procedure in the same way three more times, for each category, using the following expressions in the **Calculate** field:

`*0 + (ds2.mhw > 1) * 1`
with the file **MHW_cat2.csv**

`*0 + (ds2.mhw > 2) * 1`
with the file **MHW_cat3.csv**

`*0 + (ds2.mhw > 3) * 1`
with the file **MHW_cat4.csv**

The created **SUBSETs** are not saved as new NetCDF files.

As a result, we currently have the following files:

NetCDF files:

- **MS_sst.nc** — variable: **sst** (in degrees Celsius)
- **MS_anomaly.nc** — variable: **anomaly** (day-of-year SST anomaly)
- **MS_MHW.nc** — variable: **mhw** (categories 1, 2, 3, 4; 0 indicates no MHW)
- **MS_object.nc** — variable: **region_ID** (region object ID)

Text files:

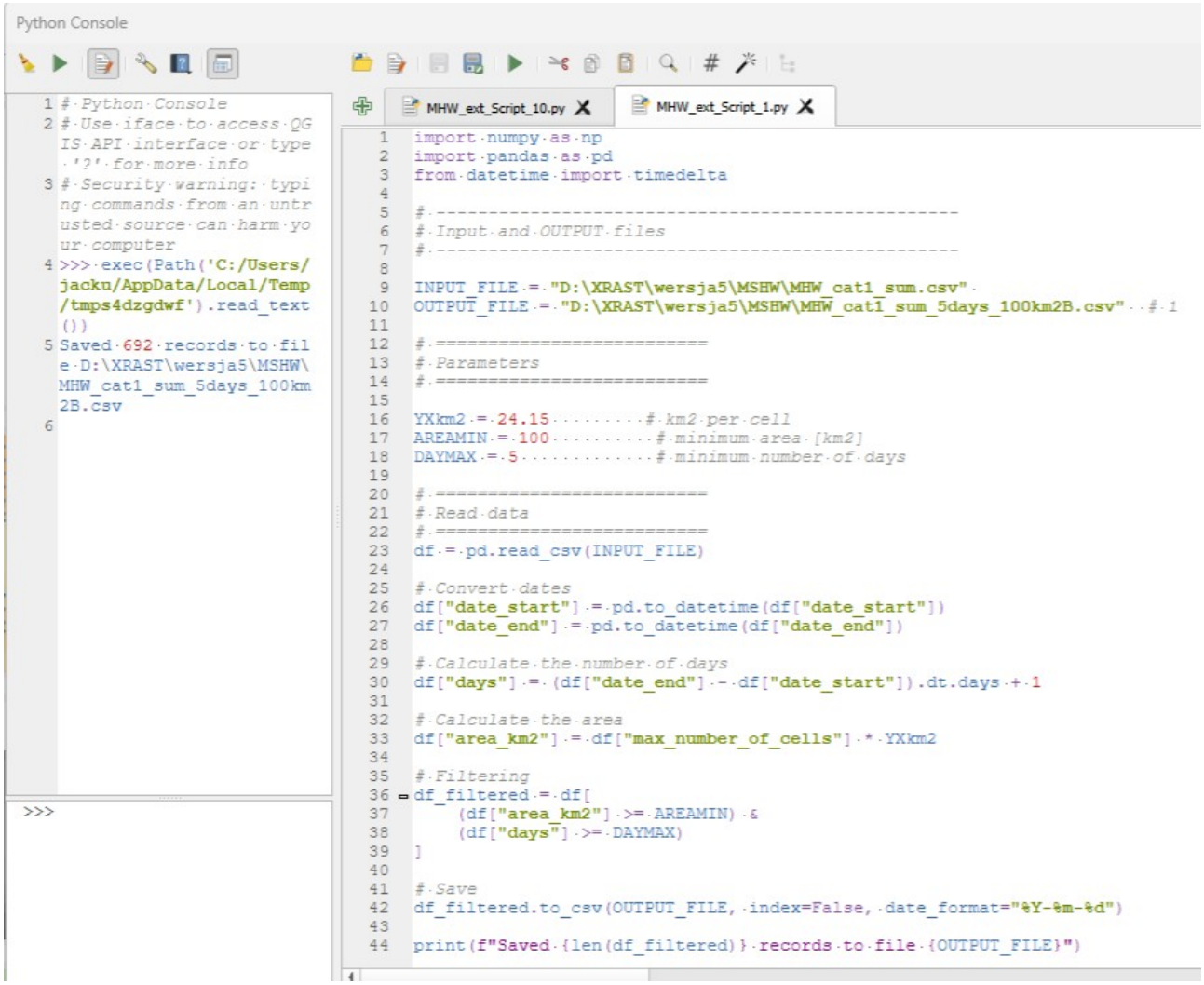
- **MHW_cat1_sum.csv**

- **MHW_cat2_sum.csv**
- **MHW_cat3_sum.csv**
- **MHW_cat4_sum.csv**

We will now create the MS_objectFFF.nc file, which will contain a mask for the region_ID objects, classifying them as MHWs according to the definition, using the required minimum duration and minimum acceptable area.

As the first step, we perform **Filtering by minimum area (km²) and duration (days)**.

This is done using the **MHW_ext_Script_1.py** script in the Python Console. Alternatively, **Script 1** can be executed in the **MHW_extension.ipynb** Jupyter Notebook.



The following file will be created, sorted by the number of days the object with a given ID existed:

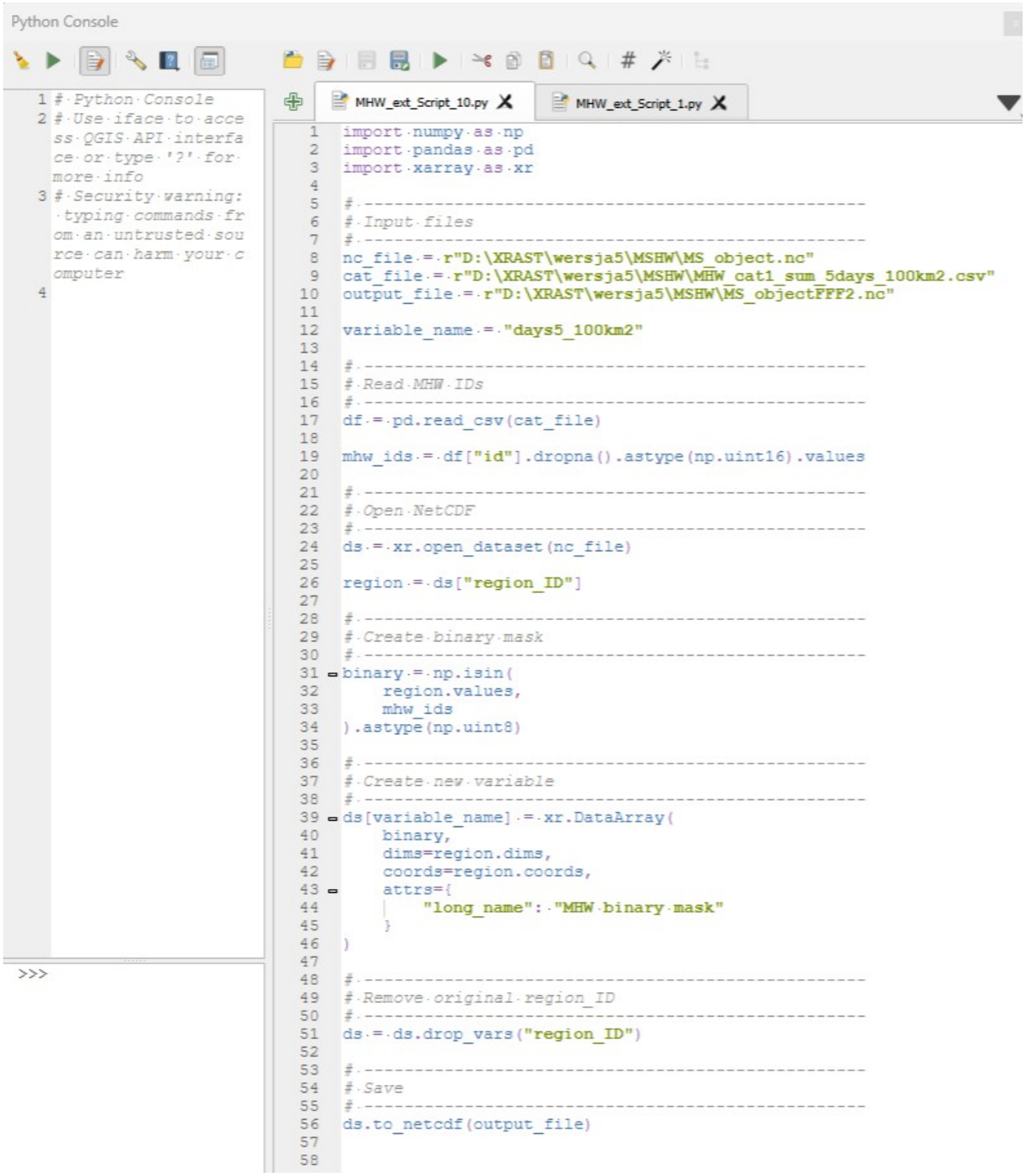
id,max_number_of_cells,date_start,date_end,date_of_max,days,area_km2
51900,6268,2024-07-08,2025-05-26,2024-07-17,323,151372.19999999998
36391,5959,2019-09-23,2020-01-03,2019-10-30,103,143909.85
657,1744,1996-12-20,1997-03-31,1997-02-11,102,42117.6
40001,3718,2020-12-11,2021-03-11,2021-02-09,91,89789.7
20741,5551,2012-09-25,2012-12-21,2012-11-27,88,134056.65
47706,6127,2023-09-27,2023-12-17,2023-10-29,82,147967.05

Next, we use the **MHW_ext_Script10.py** script in the Python Console.

This script uses the ID values from **MHW_cat1_sum_5days_100km2.csv** as a list of IDs and converts **MS_object.nc** into a binary dataset:

1. **variable_name** has a value of 1 if the original **region_ID** is present in the **id** column of **MHW_cat1_sum_5days_100km2.csv**.
2. **region_ID** has a value of 0 otherwise.
3. The same dimensions, coordinates, and time structure as in **MS_object.nc** are retained.
4. The result is saved as **MS_objectFFF.nc**.

Alternatively, Script 10 can be executed in the **MHW_extension.ipynb** Jupyter Notebook.



The **MS_objectFFF.nc** file is created, containing the **days5_100km** variable, which has a value of **1** when an MHW is present (according to the definition) and **0** when it is not present.

We will now combine the following five files:

- **MS_sst.nc** — variable: **sst** (in degrees Celsius)
- **MS_anomaly.nc** — variable: **anomaly** (day-of-year SST anomaly)

- **MS_MHW.nc** — variable: **mhw** (categories 1, 2, 3, 4; 0 indicates no MHW)
- **MS_object.nc** — variable: **region_ID** (region object ID)
- **MS_objectFFF.nc** — variable: **days5_100km** (1 when an MHW exists according to the definition, 0 otherwise)

into a single file:

MS_sst_anom_mhw_id_F.nc

containing all five variables.

We create a text file (**nc_files.csv**) containing the file paths.

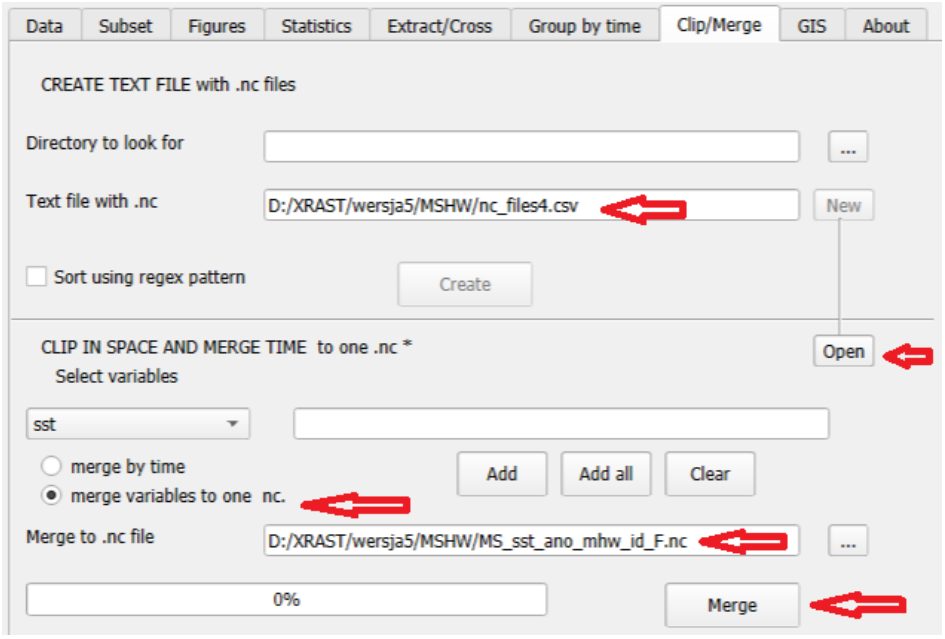
D:\XRAST\wersja5\MSHW\MS_sst.nc
D:\XRAST\wersja5\MSHW\MS_anomaly.nc
D:\XRAST\wersja5\MSHW\MS_MHW.nc
D:\XRAST\wersja5\MSHW\MS_object.nc
D:\XRAST\wersja5\MSHW\MS_objectFFF.nc

The operation of combining the files into a single file containing five variables is performed in the **Clip/Merge** tab. The settings should be entered as shown in the figure below.

Important:
All files must have the same number of rows and columns.

Before performing the operation, enter a representative subset (e.g. with the downloaded data) (**Subset → Reset → OK**) to define the **X, Y, and Z** dimensions of the output.

If the NetCDF file created by the program (after any operation) has too few rows or columns, usually by one, the **X** and **Y** ranges can be adjusted slightly in the **Subset** tab.



The created file

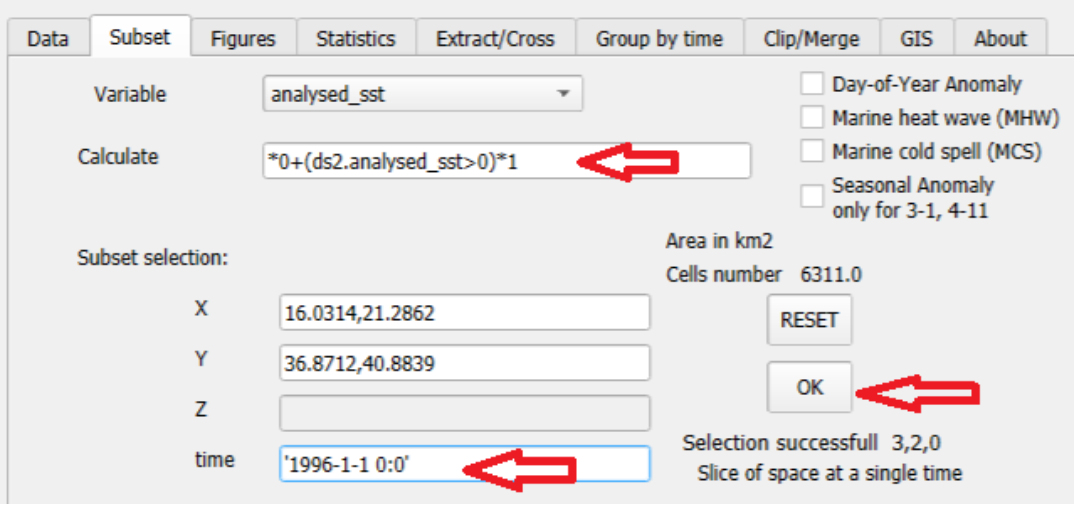
MS_sst_mhw_id_F.nc

contains five variables:

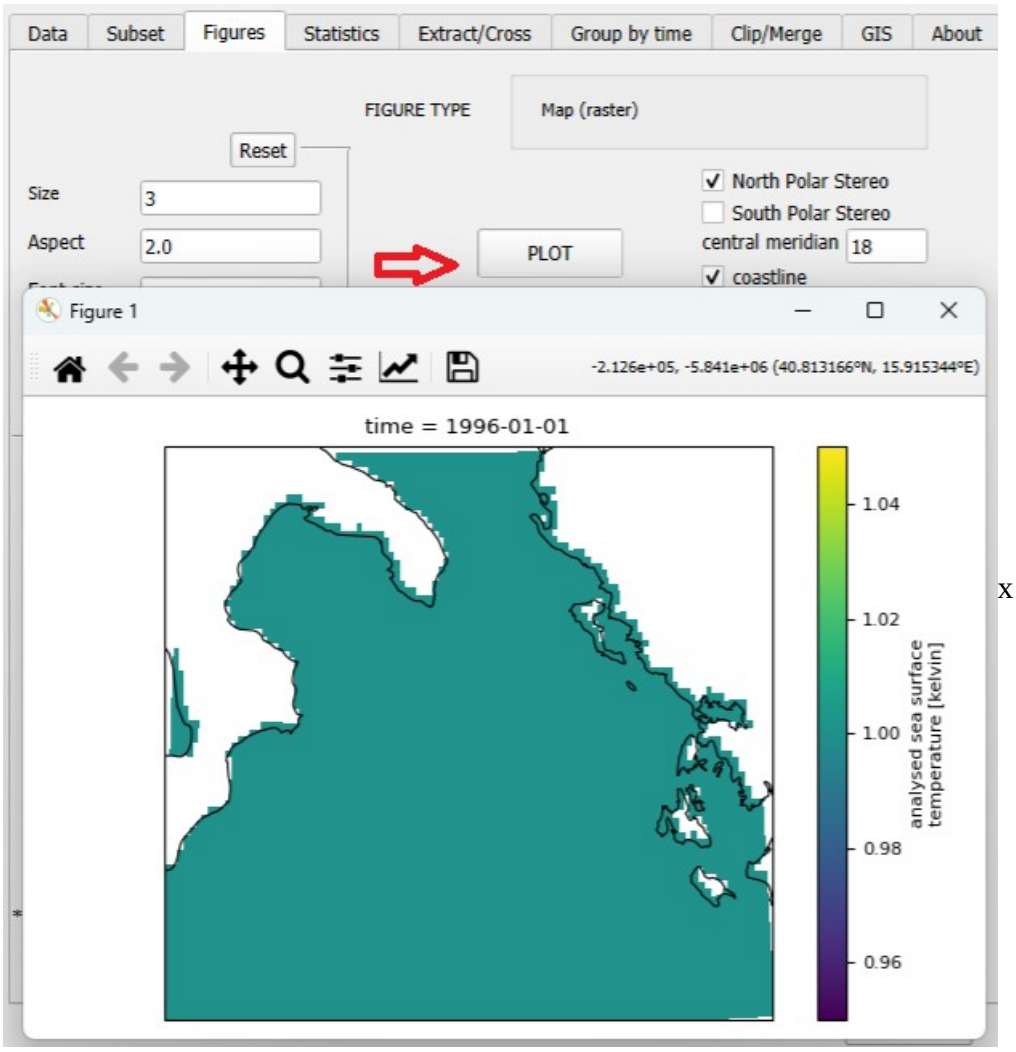
Variables:

- **sst [°C] — SST Celsius**
  SST in degrees Celsius.
- **anomaly [unknown] — unknown**
  Climatic SST anomaly (relative to the mean value in each cell for a given day of the year). This means that the baseline used to calculate the anomaly is flat.
- **mhw [unknown] — unknown**
  MHW category, determined by how much the sea temperature exceeds the **90th percentile threshold** relative to the local climatology. The categories should not be consideredequivalent to MHWs according to the Hobday definition because they do not include the requirement that the event persist continuously for at least 5 days.
- **region_ID [unknown] — unknown**
  ID of the MHW object (**region group**).
- **days5_100km2 [unknown] — MHW binary mask**
  A binary mask containing **1** when an MHW is present in a cell (according to the Hobday definition) and **0** when no MHW is present.

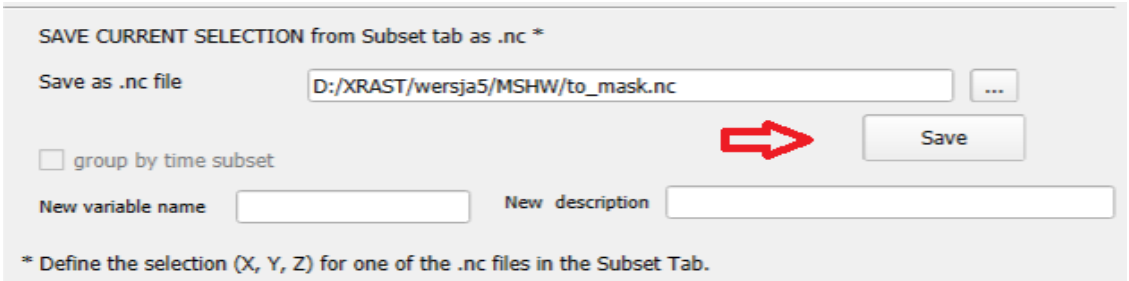
Additionally, we will create a **land–water mask** for our data. We create a single layer (for one day) containing values of **1** and **NoData** (from any available data).



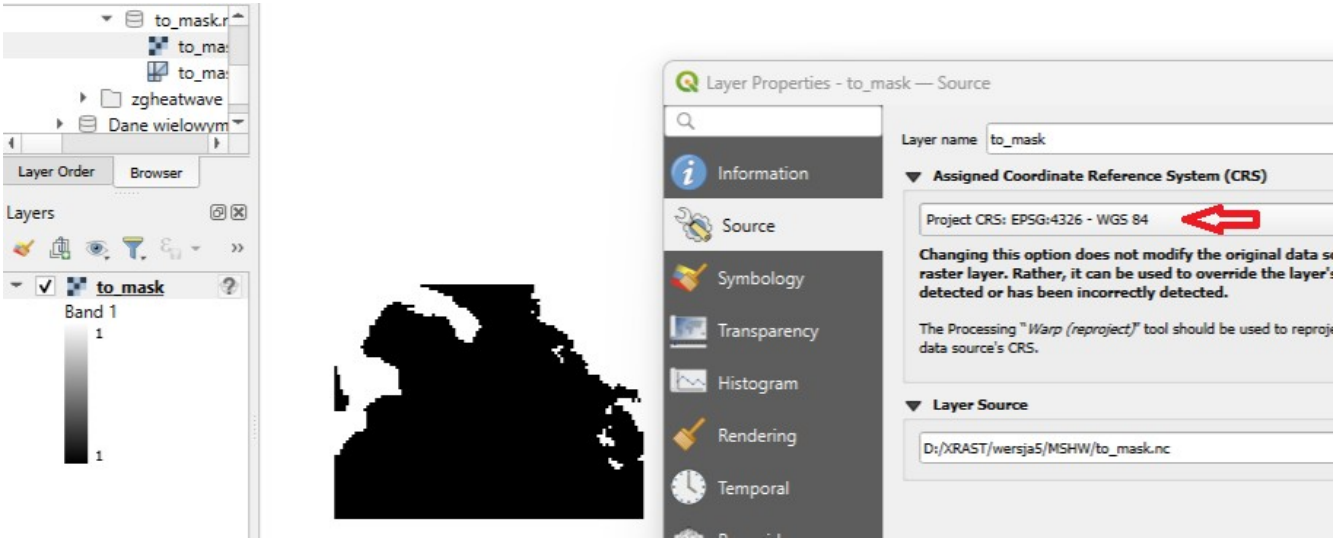
The map will have the following form:



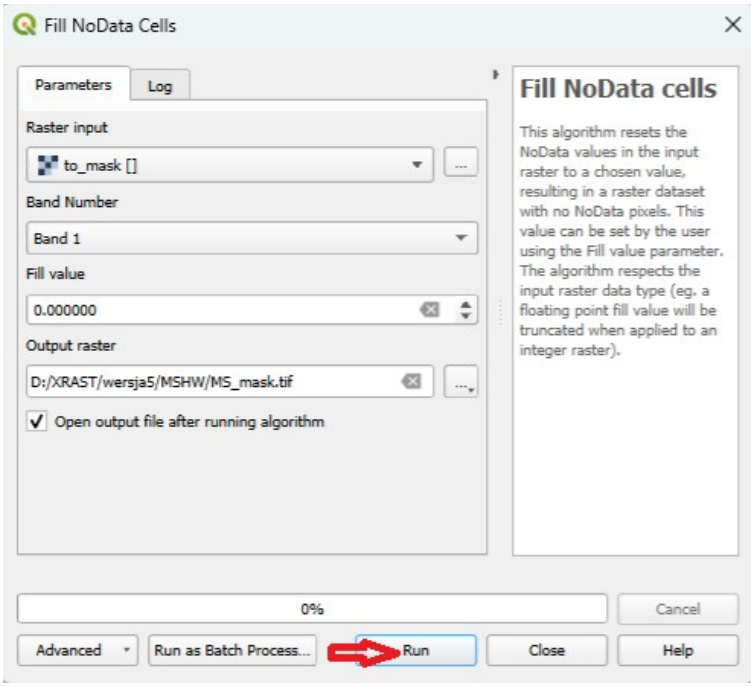
We save it as a new NetCDF file.



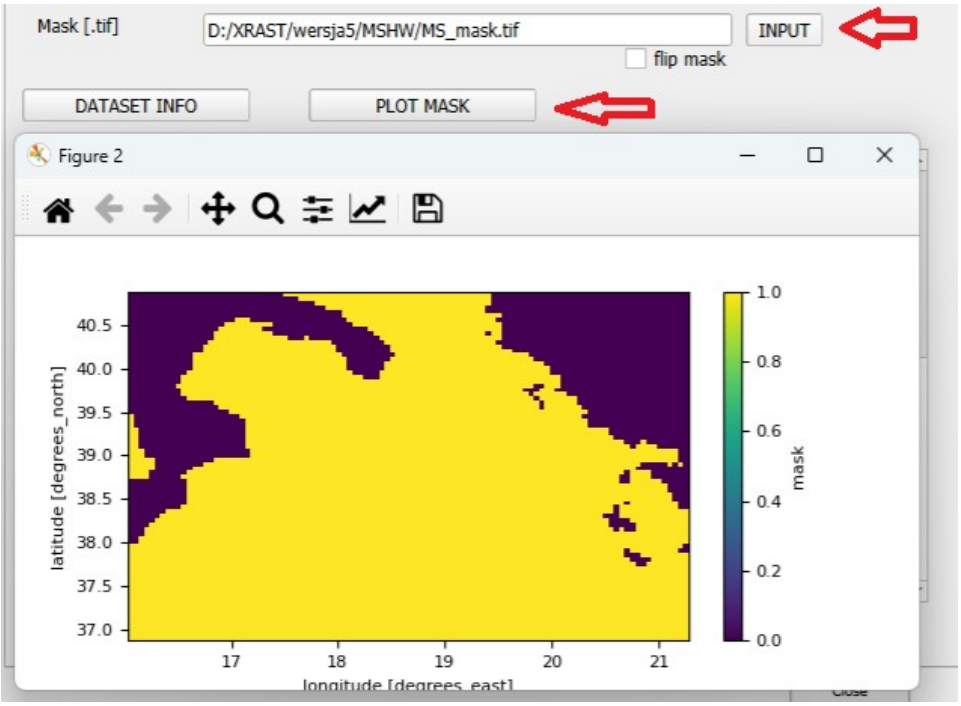
We open the file in QGIS and set the geographic coordinate reference system.



We perform the operation of filling **NoData cells** with the value **0** and save the result as a **.tif** file.



We can now use the mask in **Xraster**.



The **MS_sst_mhw_id_F.nc** (1.8 GB) **dataset can be downloaded from**: .....

https://drive.google.com/file/d/1WfKuAZAReWikcDWZqnCzNS6QufrD93pm/view?usp=sharing

Text files:

- **MHW_cat1_sum.csv**
- **MHW_cat2_sum.csv**
- **MHW_cat3_sum.csv**
- **MHW_cat4_sum.csv**

TIF file:

- **MS_mask.tif**

Are available for download as **MHW_MS.zip from :.....**

https://drive.google.com/file/d/1notdCiafcXt2h1yRCRNXHc2KrIFcvnzo/view?usp=sharing

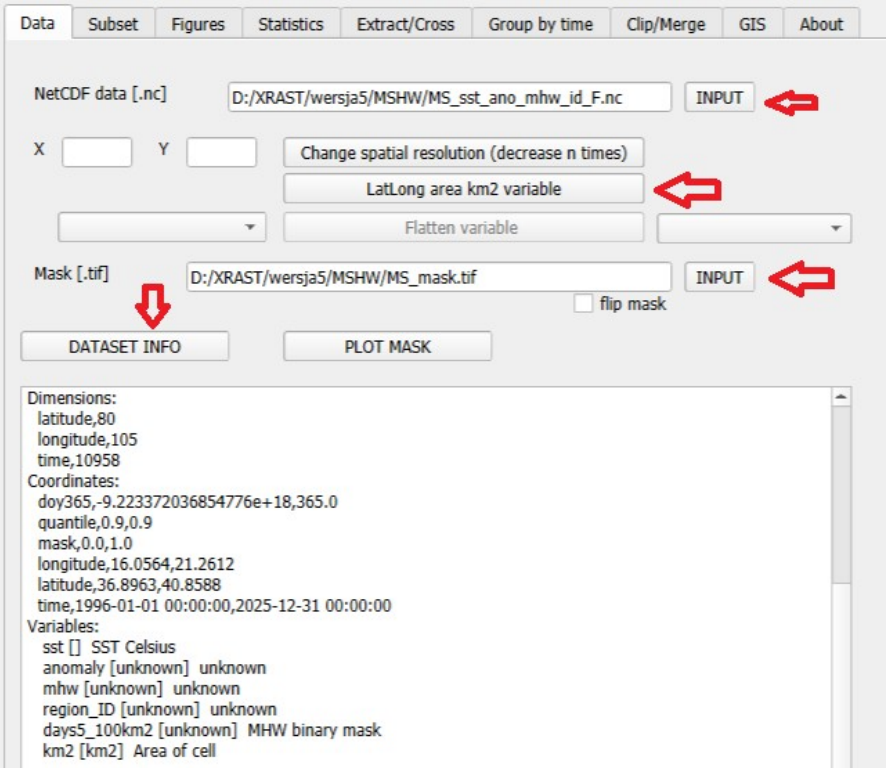
Global studies of MHWs, based on pixel values, usually include the following maps:

- **Mean annual MHW days** — map of the mean number of MHW days per year.
- **Mean intensity** — map of the mean temperature anomaly during the MHW period.
- **Maximum intensity** — map of the maximum temperature anomaly during the MHW period.
- **Cumulative intensity** — sum of temperature anomalies during the MHW period.
- **Maximum MHW category** — map of the maximum MHW category.

They also include **time-series graphs** showing how the regional or global averages and maximum values of these metrics change over time.

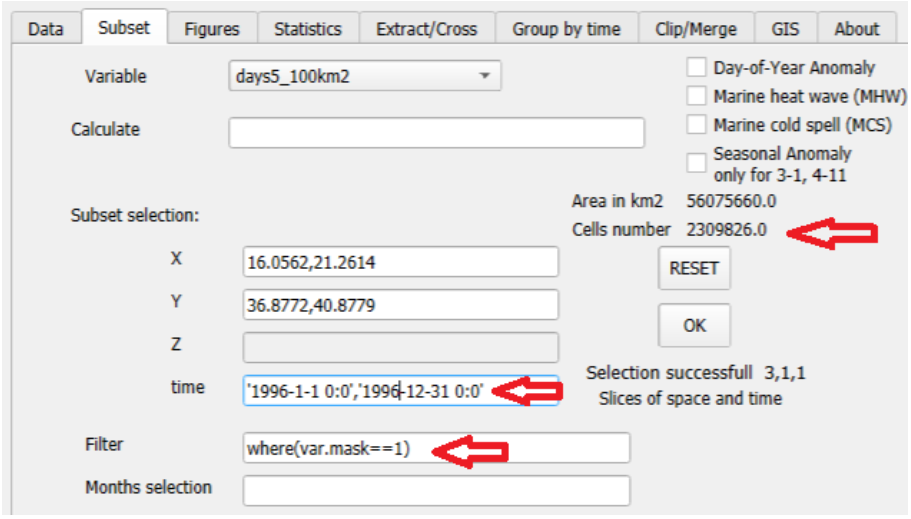
Plot of the mean number of MHW days for individual years.

In the **Data** tab, enter the data and the mask, and create the **km²** variable (using the *LatLong area km2 variable* button). After entering the data, click **DATASET INFO** to view all variables (the mask can be found under **Coordinates**).



We will use the **days5_100km2** variable (1 when MHW occurs in a cell, 0 when it does not). To calculate the annual mean, we sum the variable over the year and divide by the number of cells in the year. This gives a value from 0 to 1, representing the **MHW frequency**. To convert it to days, we multiply the value by 365.

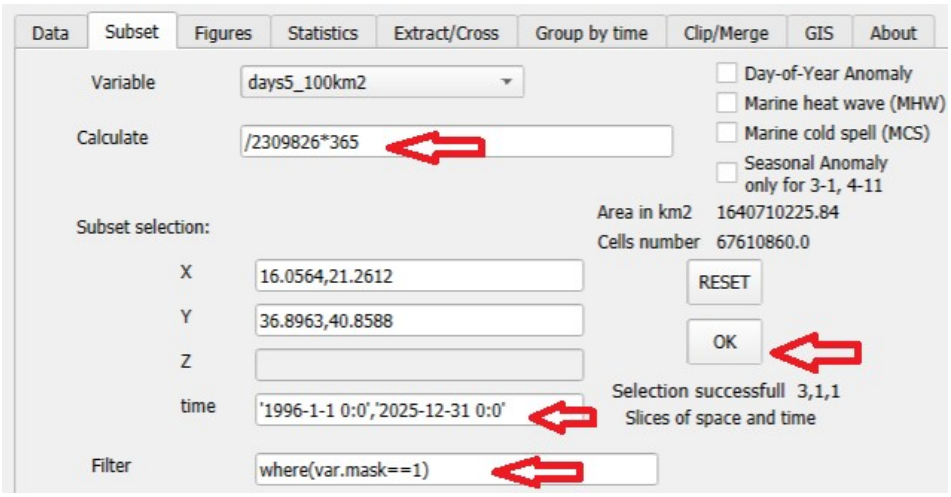
First, in the **Subset** tab, we determine the number of days per year within the area defined by the mask with a value of 1.



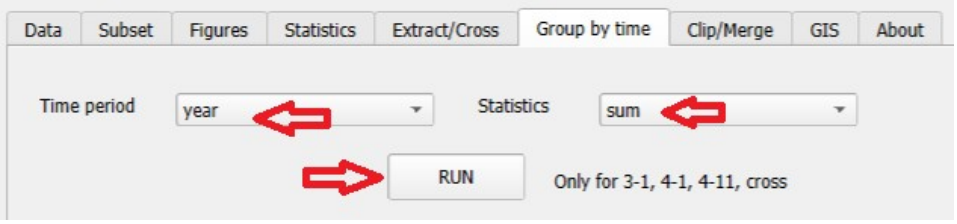
To perform the required operation, we need to assign each cell a value equal to the variable divided by the number of cells in the year (2,309,826) and multiplied by the number of days in the year.

For example, if there were 10 cells and half of them had a value of 1 (the remaining cells having a value of 0), five cells would receive the value **1/10 × 365**, while the other five would receive **0/10 × 365**. Their sum would give **182.5 mean days per year**.

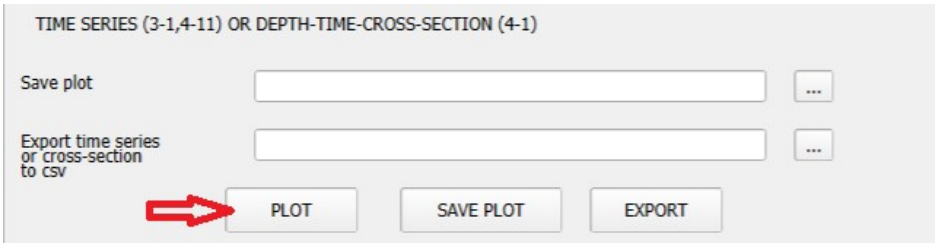
The expression in the **Calculate** field would therefore be **/2309826*365**, which is equivalent to the Map Algebra expression **variable / 2309826 * 365**.



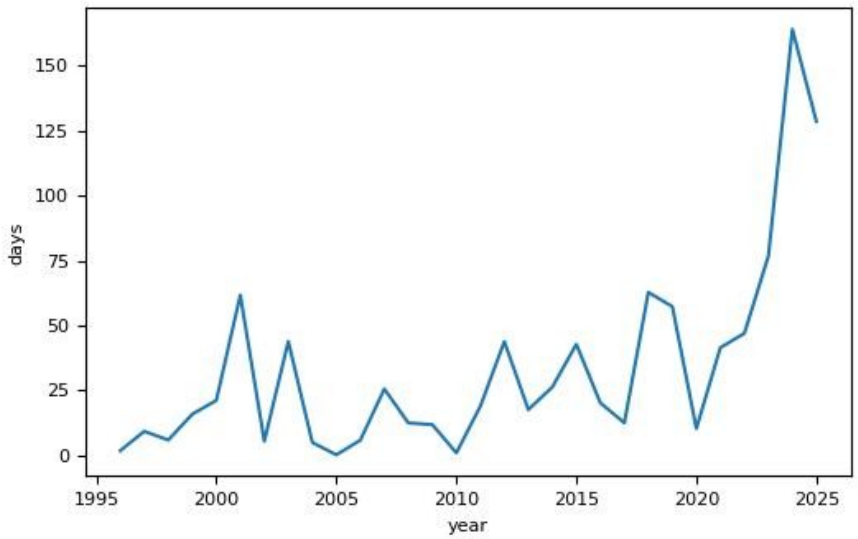
We move on to the Group by Time tab.



and click the **PLOT** button.

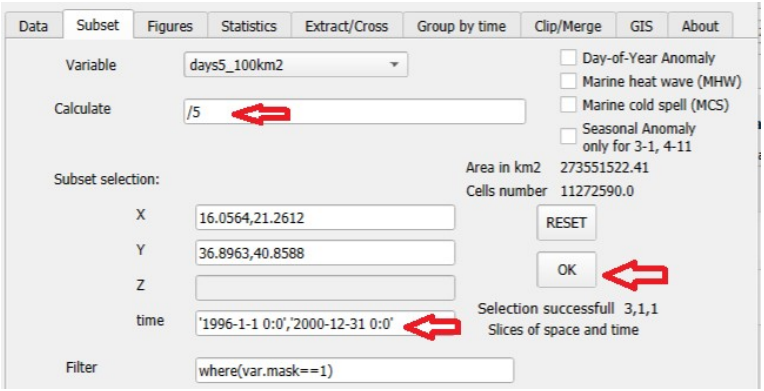


We obtain a plot showing how the number of days (spatial average) changed in individual years. The figures and maps presented have been edited using a graphics program (the editing methods are described in the manual).



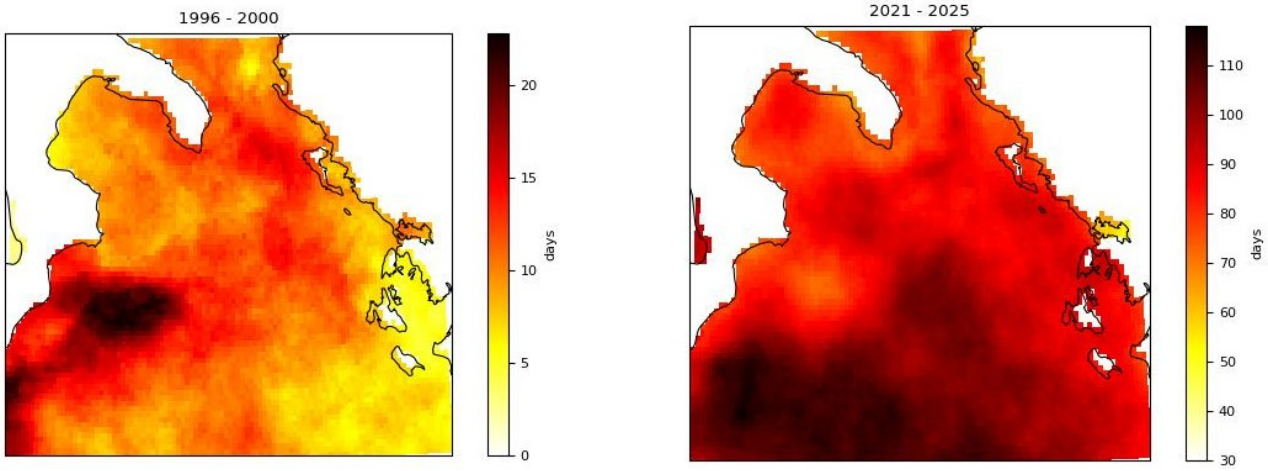
Due to the changes observed over the last three years, further mapping (mean number of MHW days, mean and maximum intensity, and cumulative intensity) will be performed for two periods: **1996–2000** and **2021–2025**. In the **Subset** tab, we create subsets of the variable divided by 5 (5 years).

Map of mean number of MHW days,



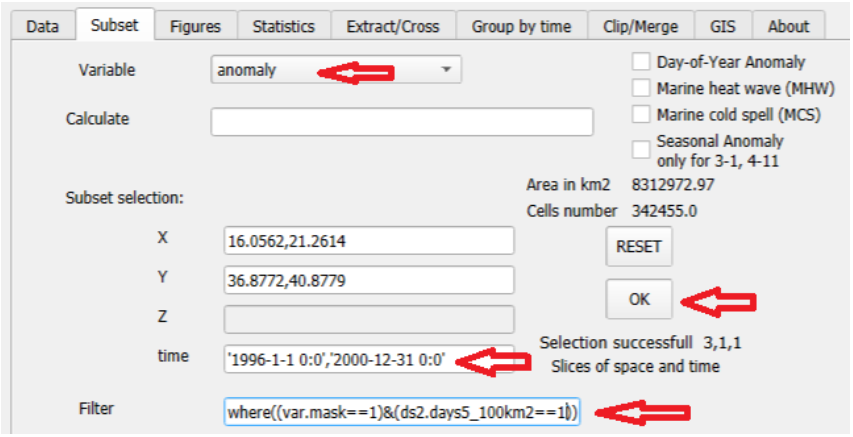
Next, in the **Statistics** tab, under the **TIME SUMMARY** panel, enter **sum** in the **Statistics** field and click **PLOT**.

We repeat these steps for both periods. We will obtain the maps.



Map of mean intensity

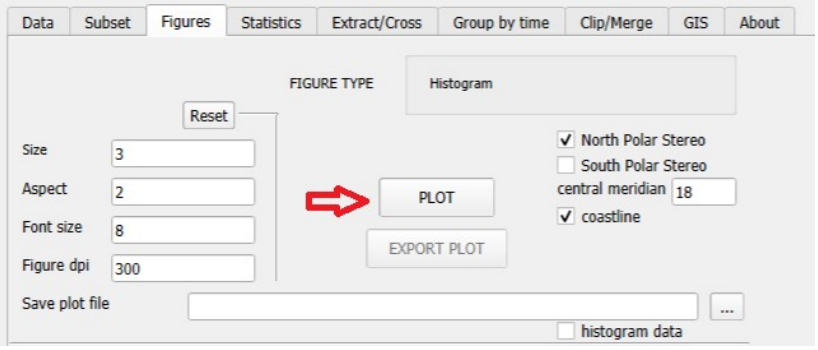
We create maps of the mean temperature anomalies during the duration of MHW events (additionally, we include histograms on the maps).



We enter the **anomaly** (intensity) variable for the first five-year period. We create a subset only for cells classified as MHW (and additionally restricted to the mask) using the following filter:

where((var.mask==1)&(ds2.days5_100km2==1))

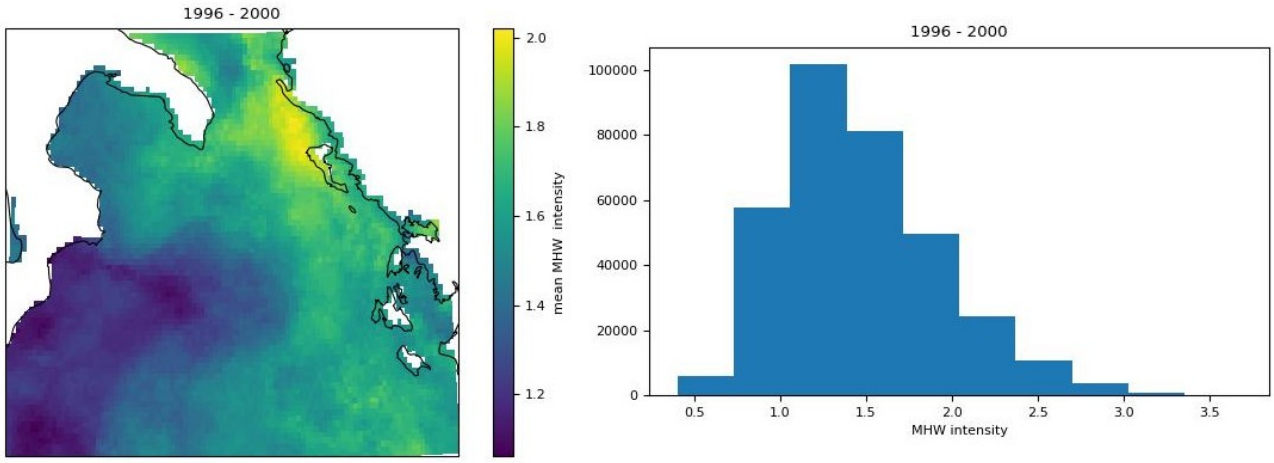
The histogram is created in the **Figures** tab. The histogram data can be saved as a text file by selecting *histogram data* and saving the plot using *Save plot file*.

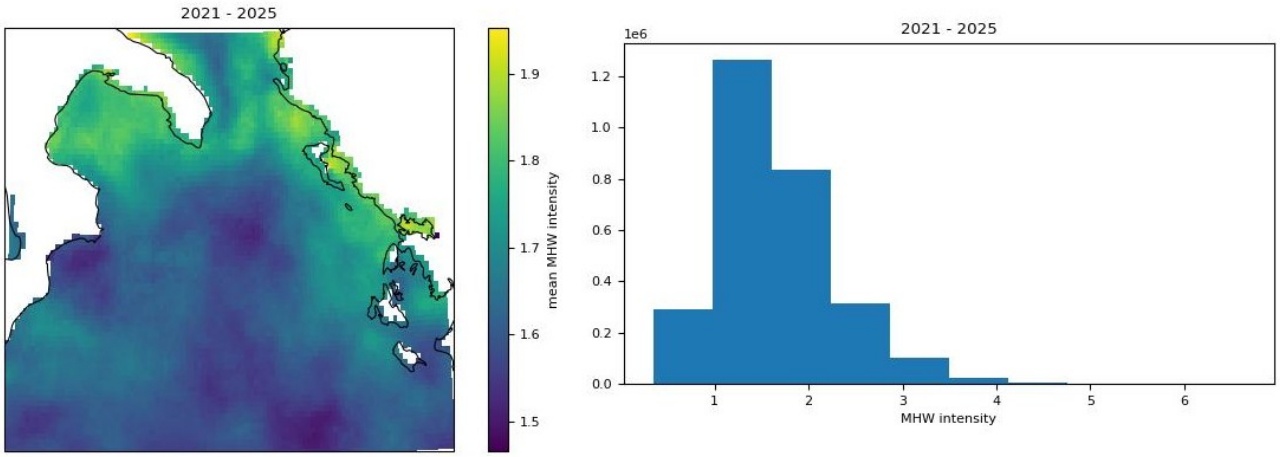


The map is created in the **Statistics** tab.



We repeat these steps for the **2021–2025** period and obtain two maps and two histograms.



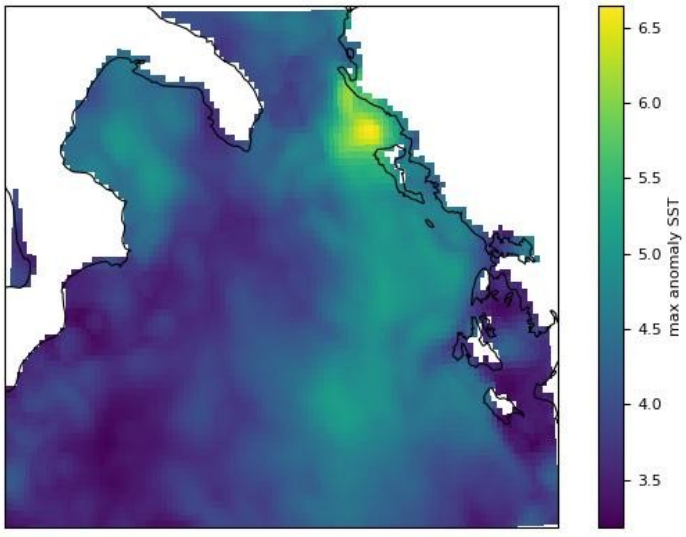


Map of maximum intensity

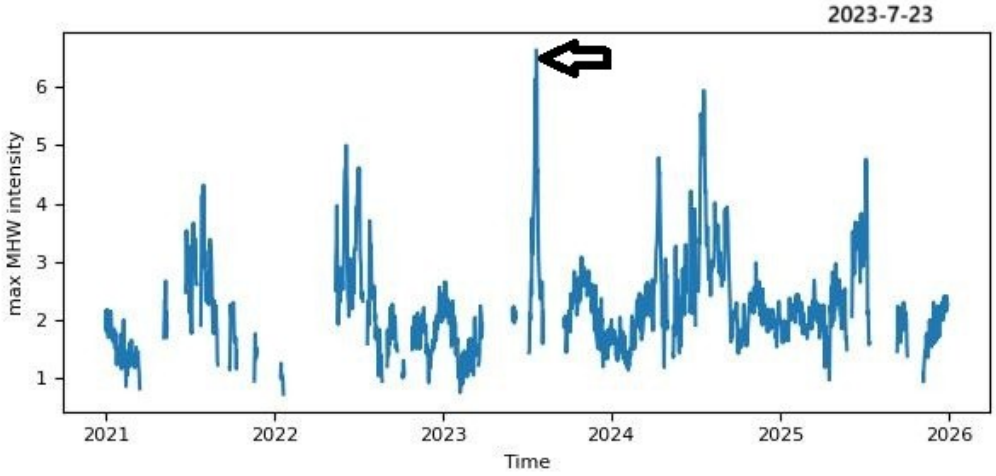
For the same subset (for the **2021–2025** period), in the **Figures** tab, we select the **max** statistic and click **PLOT**.



We obtain the map.

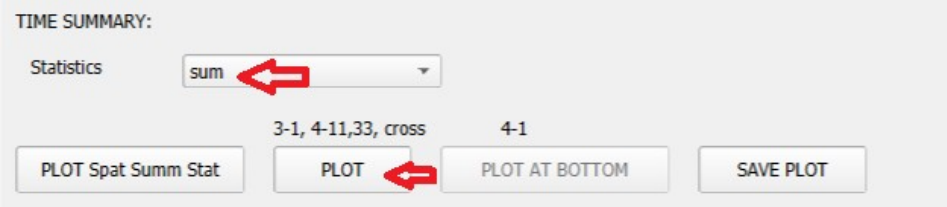


Additionally, using the **PLOT Spat Sum Stat** button allows us to determine when the maximum value occurred (using the cursor on the plot).

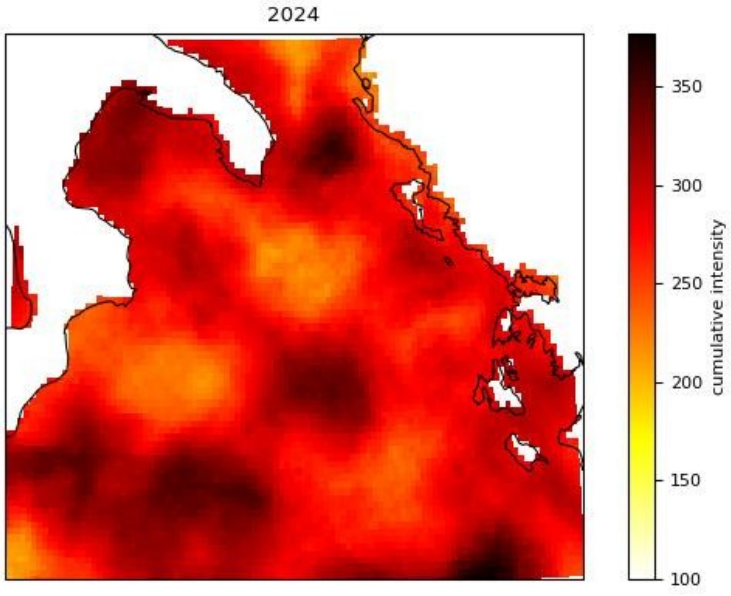


Map of cumulative intensity

The map will be created for **2024**, using the **anomaly** variable. In the **Statistics** tab, we set **Statistics** to *sum*. We sum the products of the number of days and intensity.

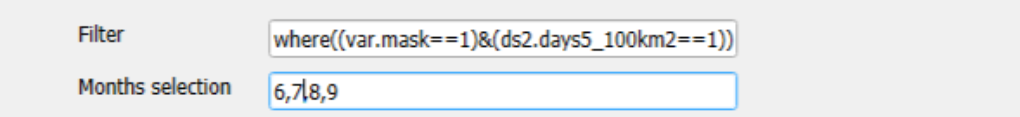


The resulting map is shown below.



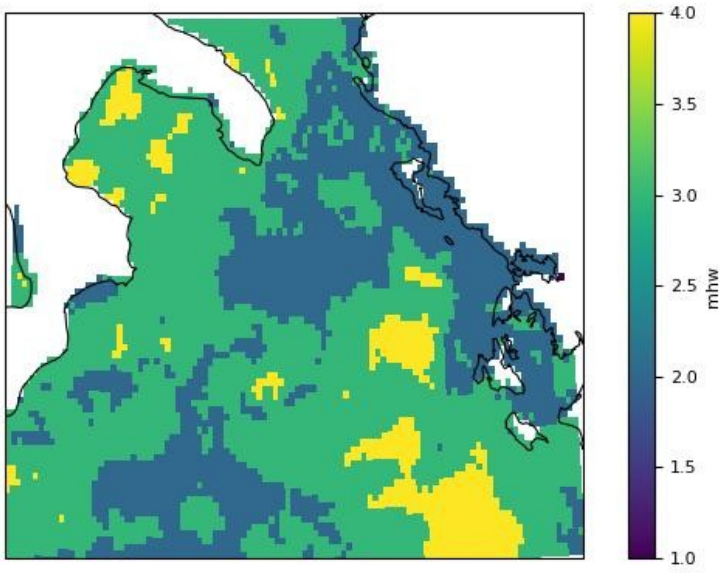
Map of maximum MHW category

The map will be created for a subset covering the entire period, but only for the summer months (June–September). When creating the subset for the MHW variable, we additionally specify the months included in the analysis.



We create the map in the **Statistics** tab by setting **Statistics** to *max*.

The resulting map will have the following form.



Object-based studies of MHW

Treat each MHW as a separate object defined in space and time.

We will use the file *"D:\XRAST\wersja5\MSHW\MHW_cat4_sum_5days_100km2.csv"*

created by filtering by minimum area (km²) and duration (days). The file describes each identified object and is sorted by duration. The file contains **693 objects** and has the following form:

id,max_number_of_cells,date_start,date_end,date_of_max,days,area_km2

51900,6268,2024-07-08,2025-05-26,2024-07-17,323,151372.19999999998

36391,5959,2019-09-23,2020-01-03,2019-10-30,103,143909.85

657,1744,1996-12-20,1997-03-31,1997-02-11,102,42117.6

40001,3718,2020-12-11,2021-03-11,2021-02-09,91,89789.7

.....

49328,4111,2024-01-13,2024-02-13,2024-01-26,32,99280.65

55049,42,2025-02-18,2025-03-21,2025-02-21,32,1014.3

25334,42,2014-11-27,2014-12-27,2014-11-29,31,1014.3

44675,2056,2022-11-03,2022-12-03,2022-11-20,31,49652.4

28326,471,2015-11-29,2015-12-28,2015-12-01,30,11374.65

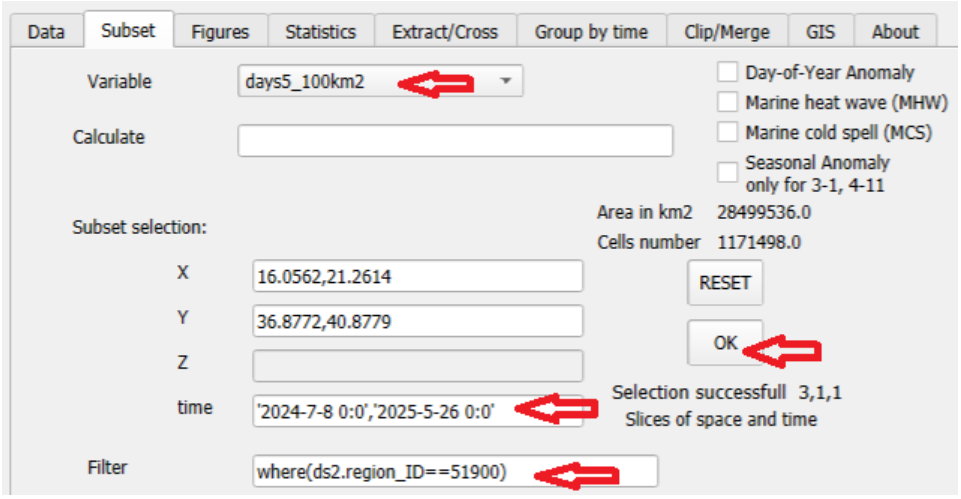
47244,6268,2023-07-09,2023-08-07,2023-07-21,30,151372.19999999998

.....

We will analyze two objects: the longest-lasting object and a significant object during the summer period.

We will start with a map showing the number of days for which object ID=51900 existed.

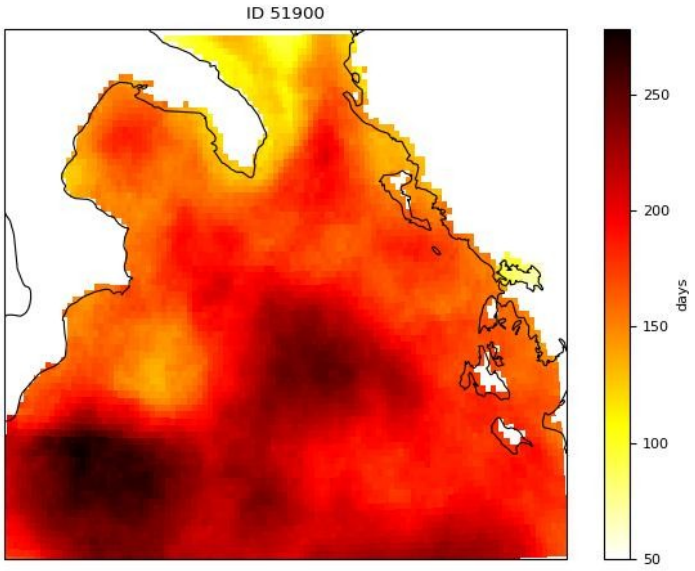
We use the `days5_100km2` variable, enter the time range (from the text file), and limit the subset to the selected object ID.



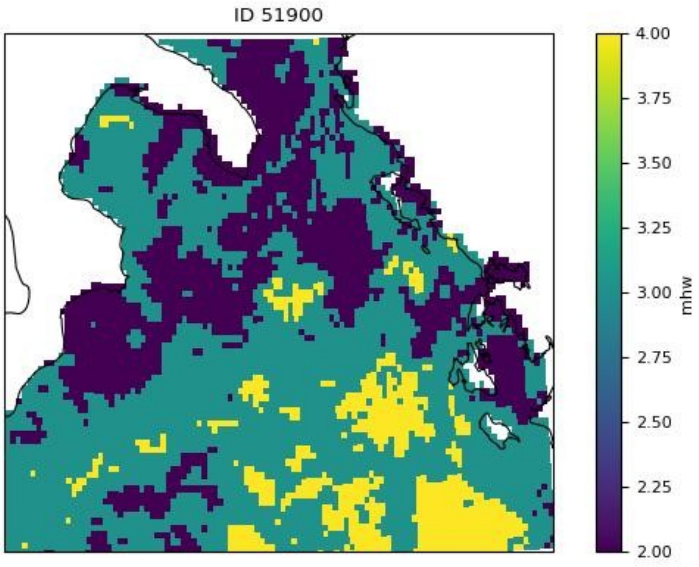
We switch to the **Statistics** tab.



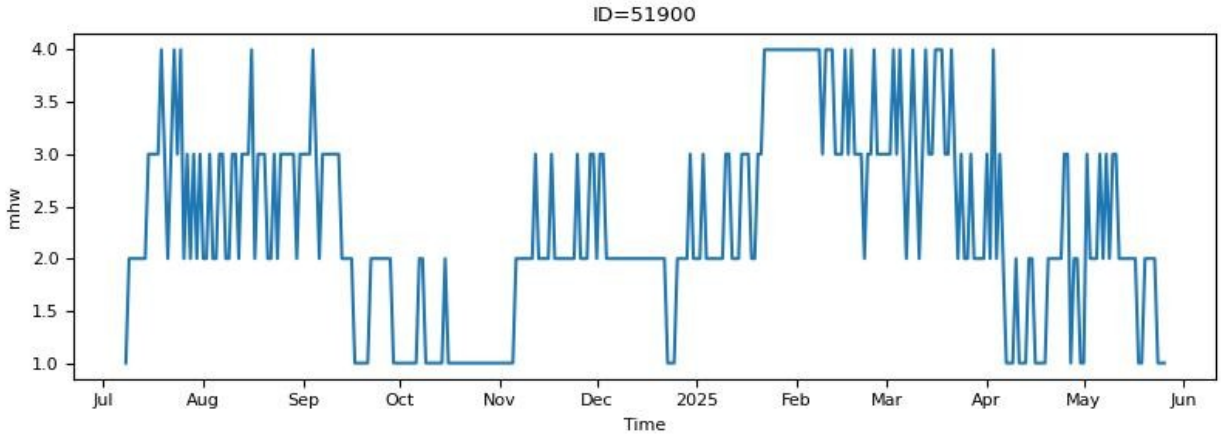
And we obtain the map.



By changing the variable to **mhw** and the statistic to **max**, we can create a map of the maximum MHW categories.

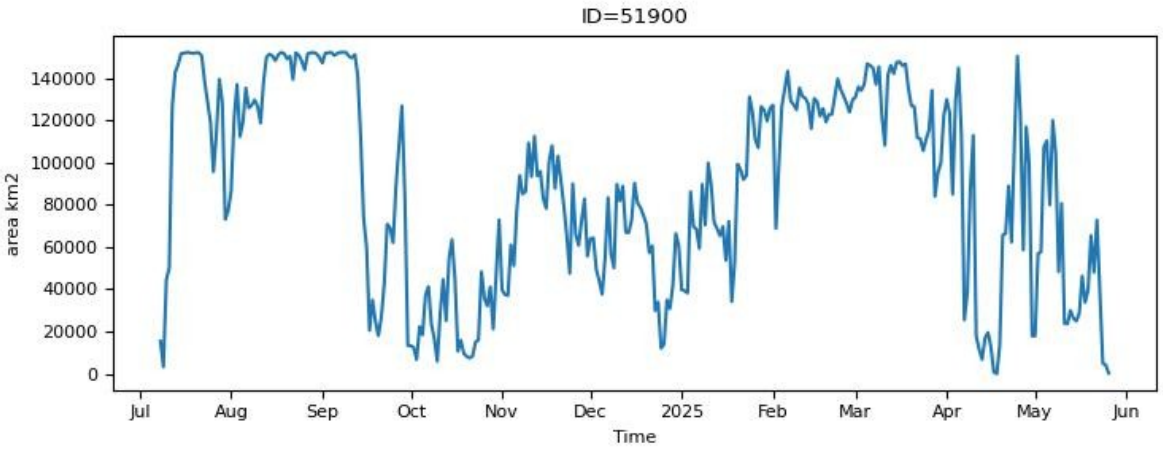


Then, by clicking **PLOT Spat Summ Stat**, we obtain a plot of the maximum MHW categories on successive days.



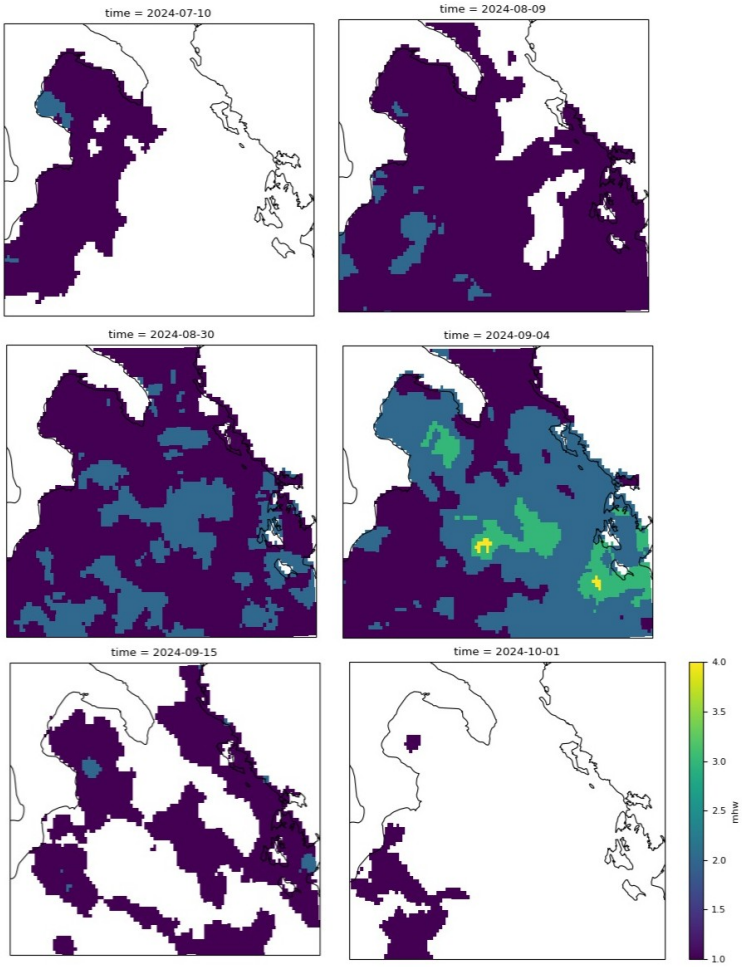
To create a plot showing changes in the MHW area for object ID=51900, we enter the variable `km2` in the **Calculate** field in the following form: (first multiplying any variable by zero).

In the **Statistics** tab, we set *statistics* to **sum** and click **PLOT SpatSumm Stat**.



We will create six maps for significant moments of change in the MHW area shown in the plot to illustrate the evolution of the MHW object over time.

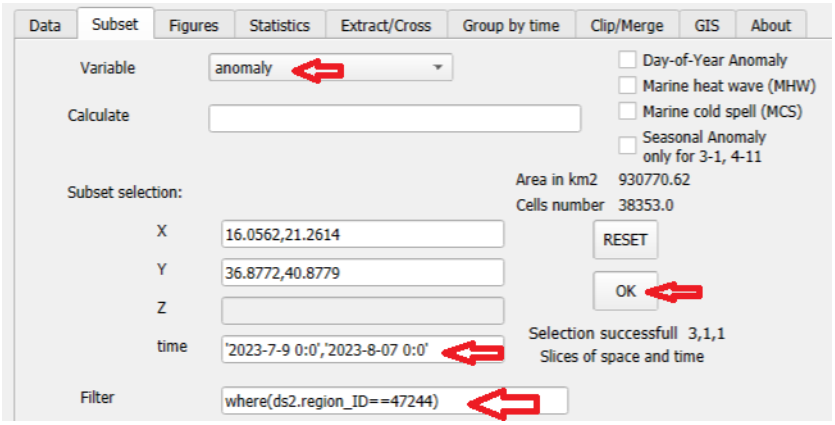
For each moment, we create a separate map of the `mhw` categories and combine them using external software. The evolution of the MHW during the first phase (until the beginning of October) is shown below.



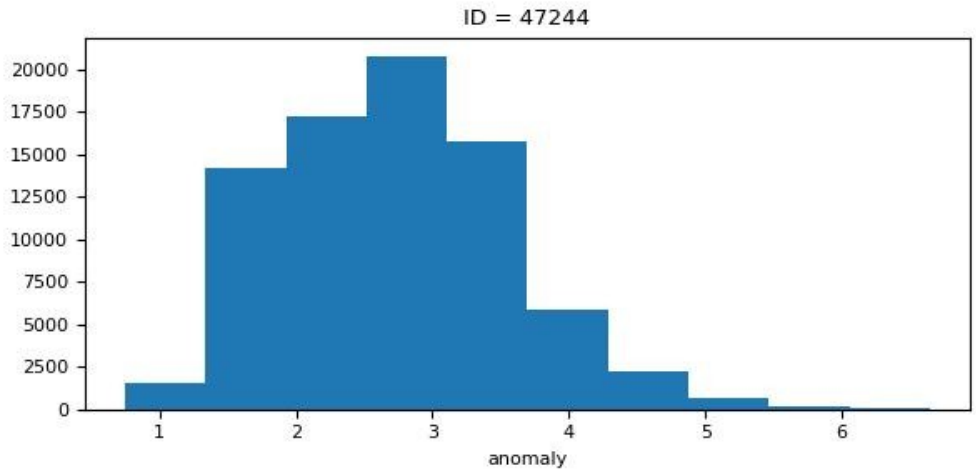
The basic parameters **mean intensity, maximum intensity, and cumulative intensity** for a specific object are determined as follows. We will analyze the following object:

47244,6268,2023-07-09,2023-08-07,2023-07-21,30,151372.19999999998

We create a subset for the `anomaly` variable and the selected object (ID and time range).



In the **Figures** tab, by clicking **PLOT**, we can create a histogram of the anomaly values for the analyzed MHW object.



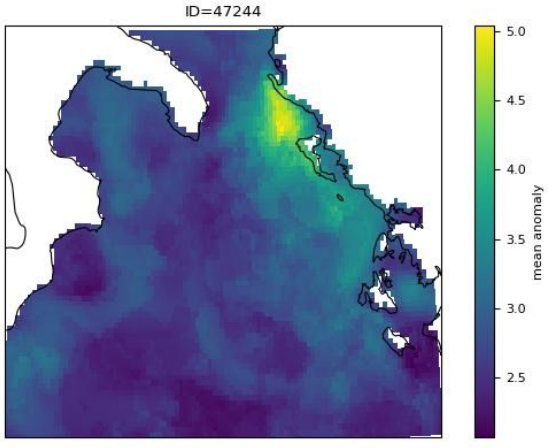
In the **Statistics** tab, we have the following options:

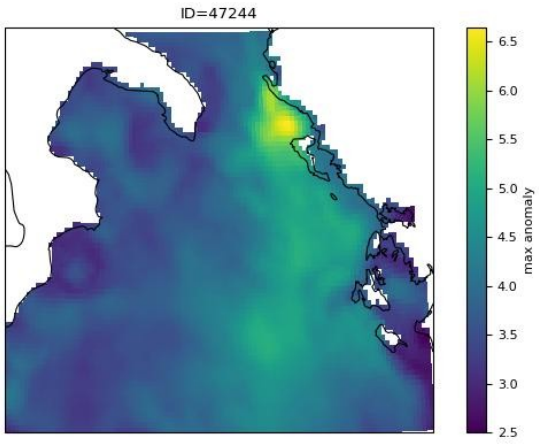
1. **Map of mean, maximum, and cumulative MHW intensity for ID=47244** — spatial distribution, with statistics calculated over time for each grid cell.
2. **Time series of mean and maximum MHW intensity for ID=47244** — temporal changes, with statistics calculated over the area of the group.
3. **A text file generated by the GIS Summary Statistics function**, providing a complete set of statistics for each day, calculated over the area of the group.

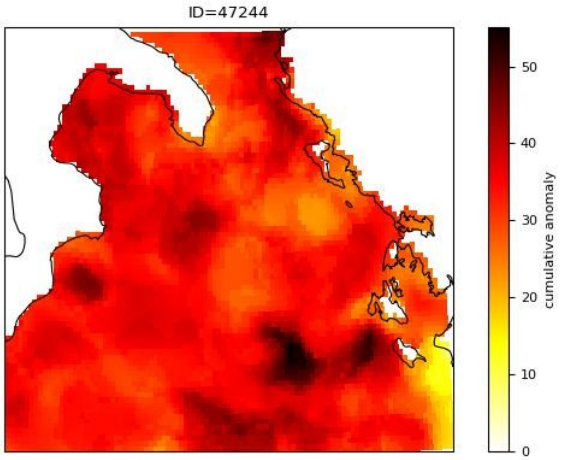
1. Map of mean, maximum, and cumulative MHW intensity for ID=47244 — spatial distribution, with statistics calculated over time for each grid cell.

In the **Statistics** tab, under **Time Summary**, we select the **mean, max, and sum** statistics and click **PLOT**.



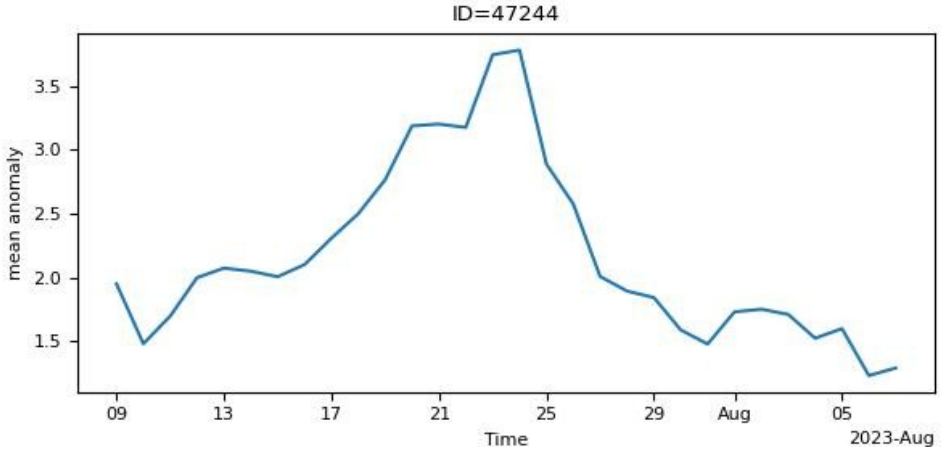


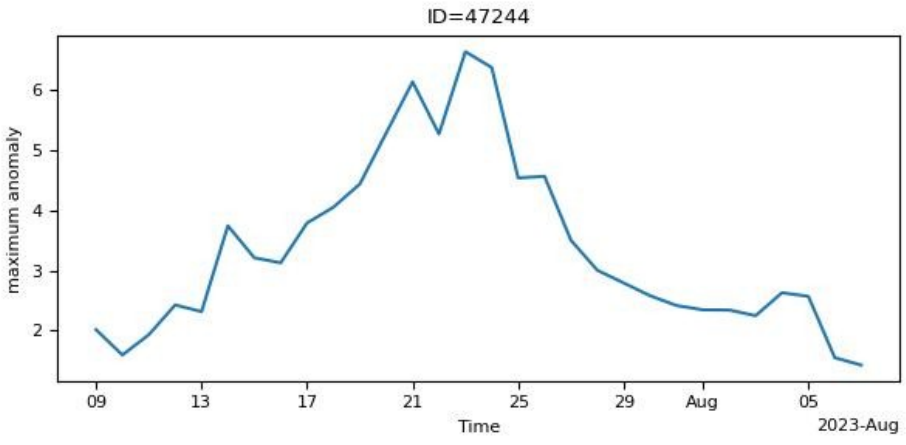




2. Time series of mean and maximum MHW intensity for ID=47244 — temporal changes, with statistics calculated over the area of the group.

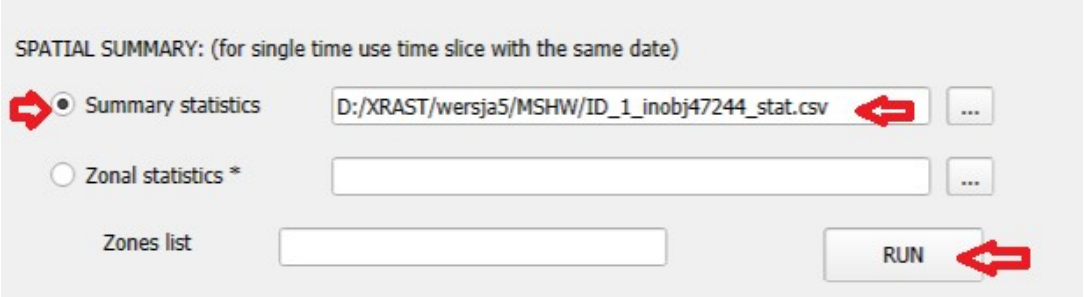
In the **Statistics** tab, under **Time Summary**, we select the **mean** and **max** statistics and click **PLOT Spat Summ Stat**.





From these figures, we can see that the **maximum intensity** occurred on 23 July. This is an important parameter describing the group.

3. A text file generated by the GIS Summary Statistics function, providing a complete set of statistics for each day, calculated over the area of the group.



The resulting file has the following format:

time,mean,min,max,median,std,sum

2023-07-09 00:00:00,1.9508,1.7887,2.0157,1.9698,0.0618, 74.13

2023-07-10 00:00:00,1.4827,1.3727,1.594,1.4832,0.0551, 44.48

2023-07-11 00:00:00,1.7201,1.3237,1.93,1.7567,0.1444, 404.22

2023-07-12 00:00:00,2.0402,1.478,2.427,2.04,0.2271, 289.71

2023-07-13 00:00:00,2.1035,1.79,2.3163,2.1163,0.1289, 250.31

2023-07-14 00:00:00,2.1132,1.281,3.7447,2.0747,0.3806, 4617.39

2023-07-15 00:00:00,1.9538,1.1197,3.212,1.9373,0.3965, 6910.69

2023-07-16 00:00:00,2.0621,1.3667,3.1283,2.0435,0.3259, 7324.73

2023-07-17 00:00:00,2.276,1.3117,3.7923,2.1727,0.4379, 11136.55

2023-07-18 00:00:00,2.4036,1.292,4.054,2.3318,0.5749, 12700.68

.....

2023-07-23 00:00:00,3.7449,1.5197,6.6433,3.7147,0.6025, 23469.1

2023-07-24 00:00:00,3.8196,2.255,6.3793,3.6917,0.6807, 23941.34

2023-07-25 00:00:00,2.9353,1.6287,4.5417,2.851,0.5139, 17382.85

2023-07-26 00:00:00,2.5204,1.326,4.5683,2.543,0.478, 13055.88

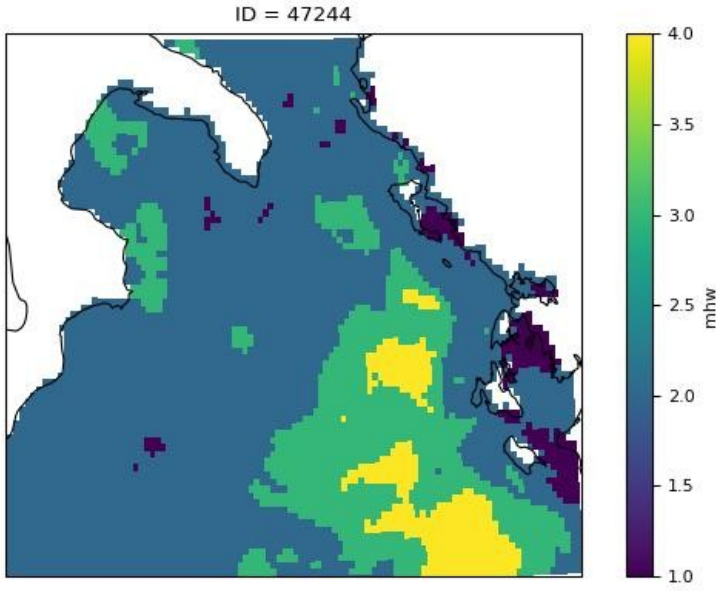
.....

The values correspond to the data used for the time series. The file can be used to calculate the **mean, maximum, and cumulative intensity** for the entire area and time period using Script **8** from ***MHW_extension.ipynb***.

The results are:

```
2.167766
6.6433
65.033
```

As before, we will create a map of the maximum MHW categories for the object with ID=47244.



The analyzed MHW contained areas belonging to four categories during its lifetime. We will examine how the area of these regions changed over time. To do this, we need to perform a **Summary Statistics** analysis for the area of each category.

We enter the `km2` variable using the expression: *any variable* ***0+ds2.km2**, and then enter the following conditions for each of the four categories:

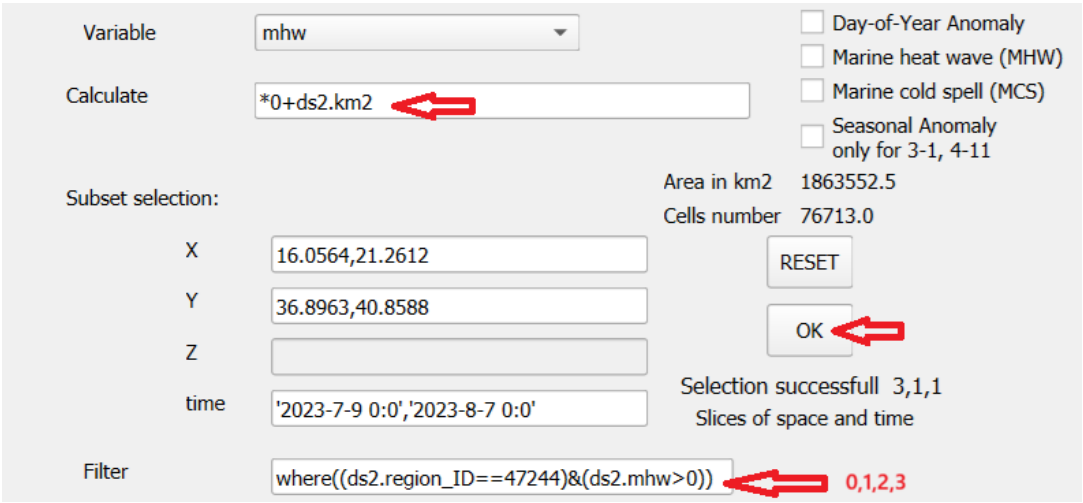
`where((ds2.region_ID==47244)&(ds2.mhw>0))` — category 1

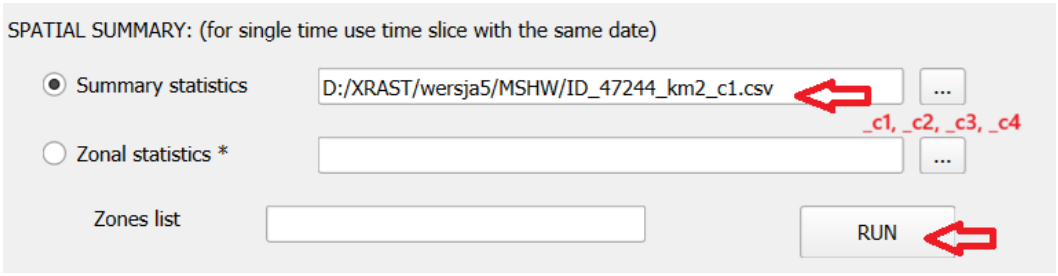
`where((ds2.region_ID==47244)&(ds2.mhw>1))` — category 2

`where((ds2.region_ID==47244)&(ds2.mhw>2))` — category 3

`where((ds2.region_ID==47244)&(ds2.mhw>3))` — category 4

In **Summary Statistics**, we create four separate text files, one for each category.





Four files were created:

ID_47244_km2_c1.csv for
`where((ds2.region_ID==47244)&(ds2.mhw>0))`

ID_47244_km2_c2.csv for
`where((ds2.region_ID==47244)&(ds2.mhw>1))`

ID_47244_km2_c3.csv for
`where((ds2.region_ID==47244)&(ds2.mhw>2))`

ID_47244_km2_c4.csv for
`where((ds2.region_ID==47244)&(ds2.mhw>3))`

For example, the file **ID_47244_km2_c3.csv** will have the following format:

time,mean,min,max,median,std,sum

2023-07-09 00:00:00,nan,nan,nan,nan,nan, 0.0

2023-07-10 00:00:00,nan,nan,nan,nan,nan, 0.0

.....

2023-07-15 00:00:00,nan,nan,nan,nan,nan, 0.0

2023-07-16 00:00:00,nan,nan,nan,nan,nan, 0.0

2023-07-17 00:00:00,nan,nan,nan,nan,nan, 0.0

2023-07-18 00:00:00,nan,nan,nan,nan,nan, 0.0

2023-07-19 00:00:00,23.7717,23.4756,24.0187,23.7397,0.1417, 2567.34

2023-07-20 00:00:00,24.0164,23.9489,24.0694,24.0187,0.0338, 720.49

2023-07-21 00:00:00,24.1032,24.1032,24.1032,24.1032,0.0, 24.1

2023-07-22 00:00:00,24.65,24.0863,24.8069,24.7245,0.2097, 4905.36

2023-07-23 00:00:00,24.4737,23.5453,24.8069,24.5089,0.2376, 29808.94

2023-07-24 00:00:00,24.5245,24.0187,24.8069,24.5597,0.193, 26143.07

2023-07-25 00:00:00,24.6612,24.3251,24.8069,24.6907,0.1128, 11294.82

2023-07-26 00:00:00,24.1359,23.7735,24.2913,24.2554,0.2123, 265.49

2023-07-27 00:00:00,nan,nan,nan,nan,nan, 0.0

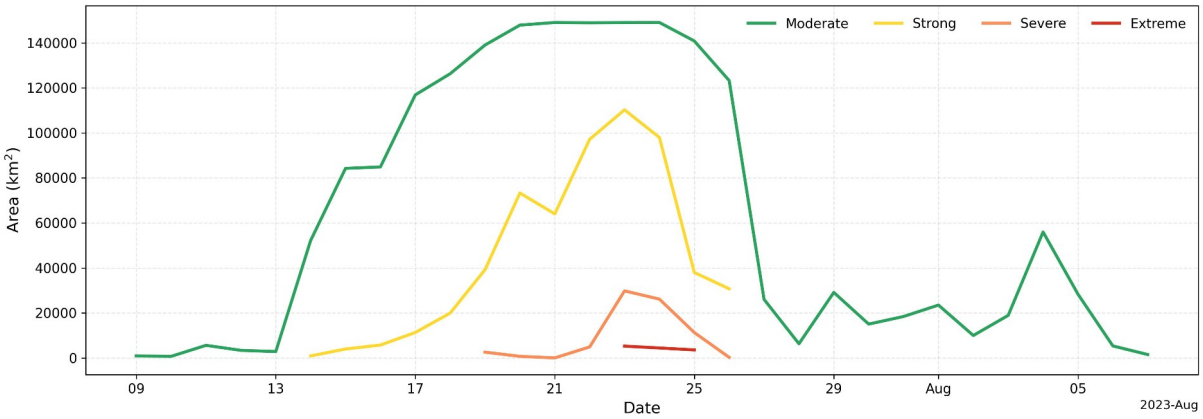
2023-07-28 00:00:00,nan,nan,nan,nan,nan, 0.0

2023-07-29 00:00:00,nan,nan,nan,nan,nan, 0.0

The values of the total area for categories 3 and 4 are recorded in the last column.

The combined plot showing changes in area will be created using **Script 7** in ***MHW_extension.ipynb***.

The resulting figure has the following form:



It describes how the area of the analyzed MHW changed over time. The plot shows that the maximum intensity, corresponding to the maximum area, was reached on 23 July.

The temporal variability of MHW, broken down by seasons, is well illustrated by bar charts showing the number of MHW days per year, created separately for each category.

We will use the filtered file:

Filtering by minimum area (km²) and duration (days)

using the `MHW_ext_Script_1.py` script in the Python Console or **Script 1** from ***MHW_extension.ipynb***:

```
INPUT_FILE = "D:\XRAST\wersja5\MSHW\MHW_cat1_sum.csv"
OUTPUT_FILE = "D:\XRAST\wersja5\MSHW\MHW_cat1_sum_5days_100km2.csv"  # 1

INPUT_FILE = "D:\XRAST\wersja5\MSHW\MHW_cat2_sum.csv"
OUTPUT_FILE = "D:\XRAST\wersja5\MSHW\MHW_cat1_sum_3days_100km2.csv"  # 2

INPUT_FILE = "D:\XRAST\wersja5\MSHW\MHW_cat3_sum.csv"
OUTPUT_FILE = "D:\XRAST\wersja5\MSHW\MHW_cat1_sum_2days_100km2.csv"  # 3

INPUT_FILE = "D:\XRAST\wersja5\MSHW\MHW_cat4_sum.csv"
OUTPUT_FILE = "D:\XRAST\wersja5\MSHW\MHW_cat1_sum_1days_50km2.csv"  # 4
```

The filtering threshold values can be specified as required.

The process is carried out by executing a sequence of scripts in ***MHW_extension.ipynb***:

Script 2 **-** ***Sort by number of days X 4***

INPUT_FILE = "D:\XRAST\wersja5\MSHW\MHW_cat1_sum_5days_100km2.csv" #1
OUTPUT_FILE = "D:\XRAST\wersja5\MSHW\MHW_cat1_sum_5days_100km2_Sdays.csv" #2

Script 4 ***- Sum of days by season***

INPUT_FILE = "D:\XRAST\wersja5\MSHW\MHW_cat1_sum_5days_100km2_Sdays.csv" # 2

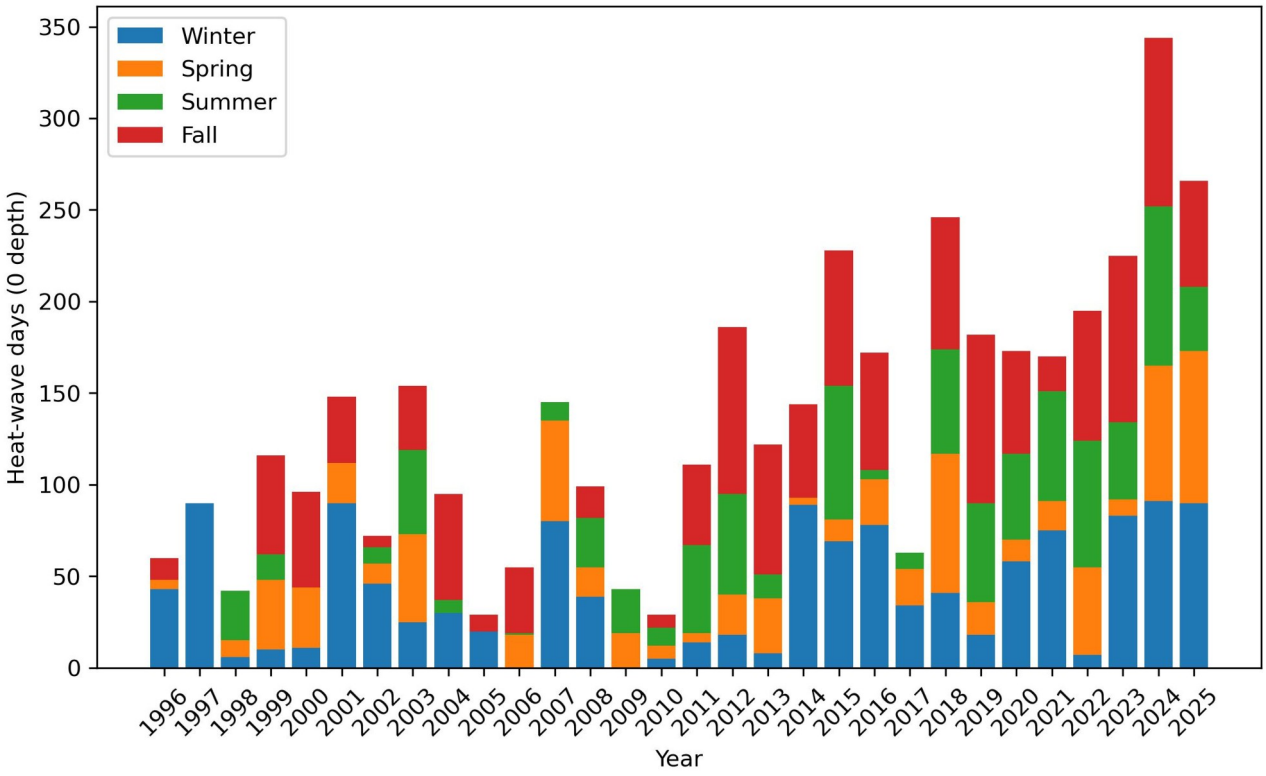
OUTPUT_FILE = "D:\XRAST\wersja5\MSHW\MHW_cat1_sum_5days_100km2_Sdays_seasons.csv" # 3

Script 5 ***- Stacked bar chart of MHW (Fig 3)***

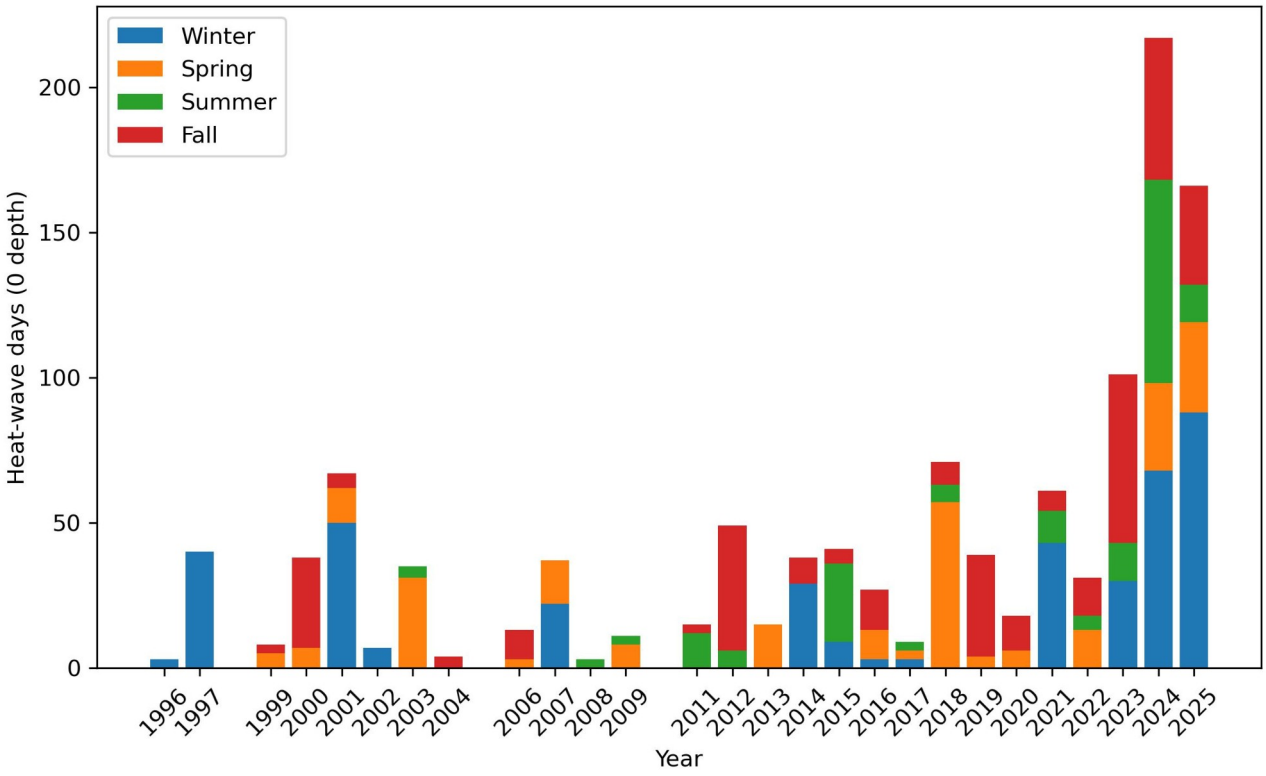
df = pd.read_csv("D:\XRAST\wersja5\MSHW\MHW_cat1_sum_5days_100km2_Sdays_seasons.csv") # 3

Four sequences are performed, one for each category. The resulting plots will have the following form

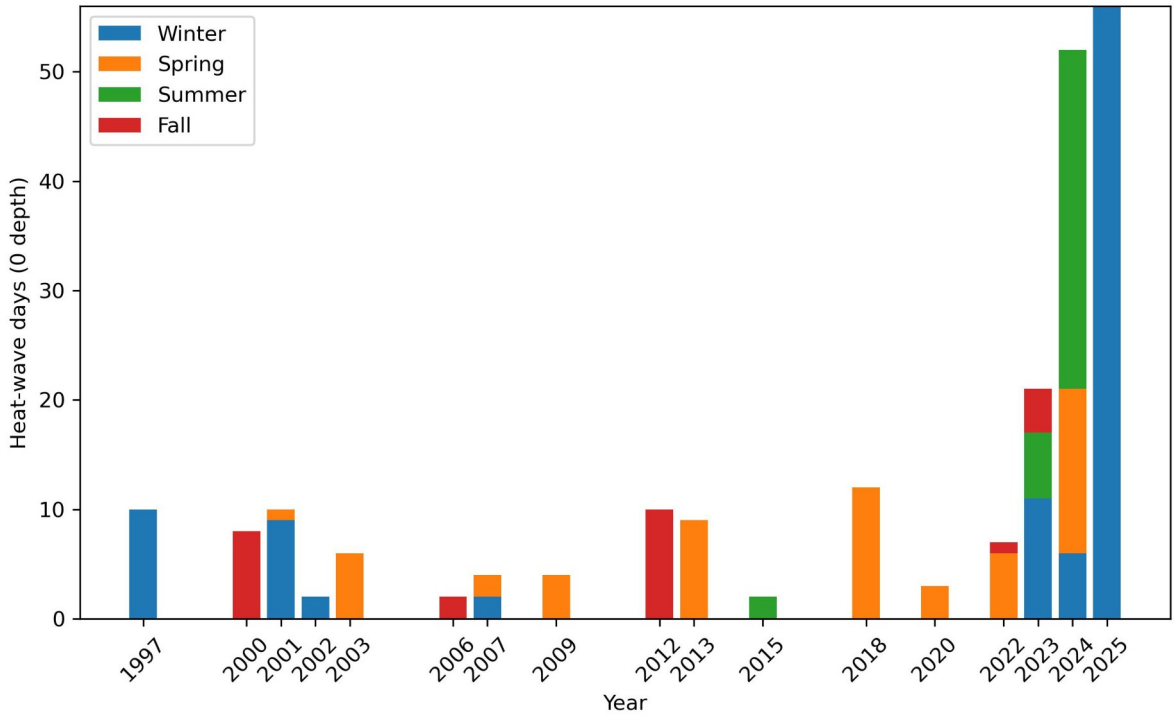
Category 1



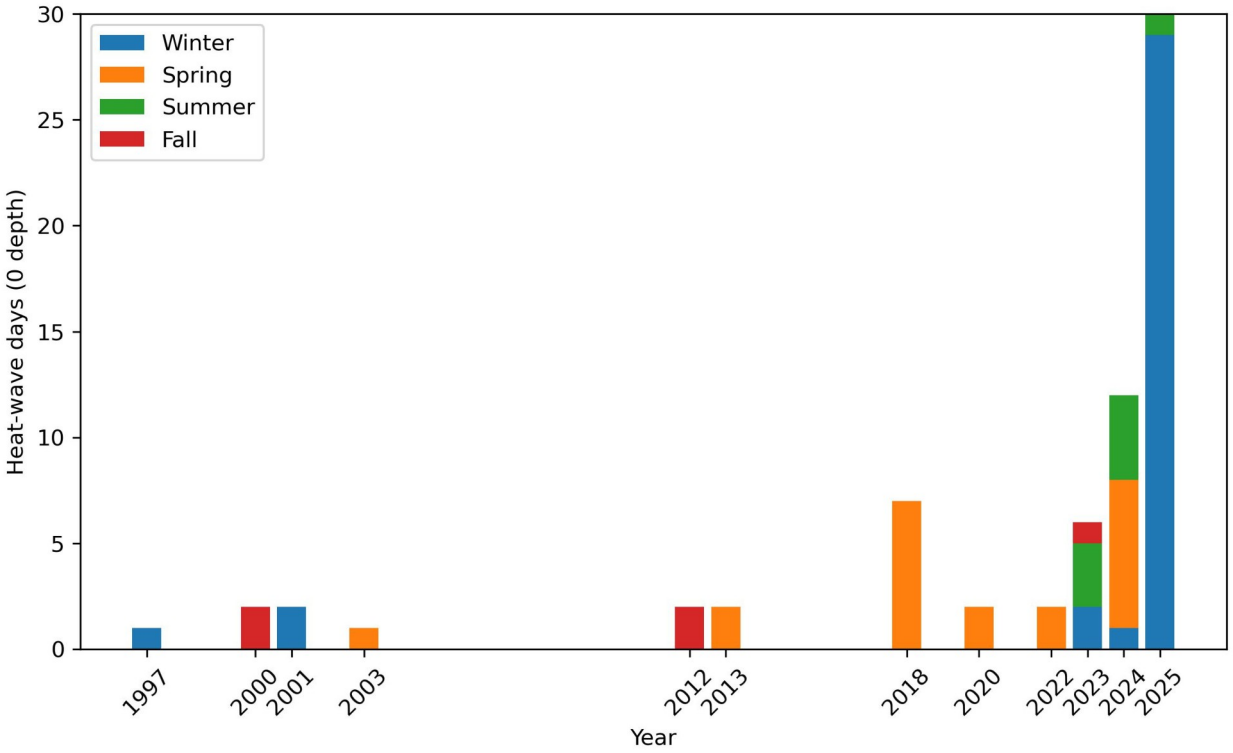
Category 2



Category 3



Category 4



The text file containing the nested groups by category (**Script 6 – Create nested groups by category**) links groups from different categories over time. It assumes that within a category 1 group, areas of category 2 could have developed, and so on; however, this is not always the case.

The file is created by adding records to a category 1 MHW record with a given identifier. Records from category 2 whose period is contained within the period of the category 1 record are added to the corresponding category 1 record. The same operation is then performed for categories 3 and 4.

This allows objects belonging to a given category to be identified. If we are looking for objects that reach at least category 2, then:

```
ff1 = "D:\XRAST\wersja5\MSHW\MHW_cat1_sum_5days_100km2.csv" # 1 for category 1
ff2 = "D:\XRAST\wersja5\MSHW\MHW_cat2_sum_3days_100km2.csv" # 1 for category 2
ff3 ="D:\XRAST\wersja5\MSHW\MHW_cat3_sum_2days_100km2.csv" # 1 for category 3
ff4 = "D:\XRAST\wersja5\MSHW\MHW_cat4_sum_1days_50km2.csv" # 1 for category 4


out_file = "D:\XRAST\wersja5\MSHW\MHW_cat1234_nested.csv" # 4
```

The file will have the following format:

```
id,max_number_of_cells,date_start,date_end,date_of_max,days,area_km2,kff
5,39,1996-01-01,1996-01-07,1996-01-04,7,941.85,1
4,1064,1996-01-01,1996-01-12,1996-01-08,12,25695.6,1
80,322,1996-01-14,1996-01-18,1996-01-16,5,7776.299999999999,1
79,94,1996-01-14,1996-01-20,1996-01-16,7,2270.1,1
```

.....
657,1744,**1996-12-20,1997-03-31**,1997-02-11,**102**,42117.6,1
397,106,1997-01-15,1997-01-17,1997-01-15,3,2559.9,2
399,43,1997-01-16,1997-01-20,1997-01-19,5,1038.45,2
417,225,1997-01-21,1997-02-09,1997-01-31,20,5433.75,2
423,18,1997-01-23,1997-01-26,1997-01-24,4,434.7,2
440,75,1997-01-30,1997-02-03,1997-01-31,5,1811.25,2
465,855,1997-02-08,1997-02-13,1997-02-11,6,20648.25,2
530,147,1997-02-25,1997-03-06,1997-03-06,10,3550.05,2
36,32,1997-01-31,1997-02-02,1997-01-31,3,772.8,3
41,26,1997-02-02,1997-02-03,1997-02-03,2,627.9,3
46,15,1997-02-05,1997-02-06,1997-02-05,2,362.25,3
55,33,1997-02-10,1997-02-13,1997-02-11,4,796.9499999999999,3
2,3,1997-02-10,1997-02-10,1997-02-10,1,72.44999999999999,4
1428,170,1998-02-24,1998-03-01,1998-02-28,6,4105.5,1
.....

The MHW object with ID=657 existed from 1996-01-30 to 1997-02-03, i.e. for 102 days. Since it contained category 4 areas, it is classified as an **extreme** MHW.

A description of this object can be saved in a separate file.

ID_1234_657.csv.

id,max_number_of_cells,date_start,date_end,date_of_max,days,area_km2,kff
657,1744,1996-12-20,1997-03-31,1997-02-11,102,42117.6,1
397,106,1997-01-15,1997-01-17,1997-01-15,3,2559.9,2
399,43,1997-01-16,1997-01-20,1997-01-19,5,1038.45,2
417,225,1997-01-21,1997-02-09,1997-01-31,20,5433.75,2
423,18,1997-01-23,1997-01-26,1997-01-24,4,434.7,2
440,75,1997-01-30,1997-02-03,1997-01-31,5,1811.25,2
465,855,1997-02-08,1997-02-13,1997-02-11,6,20648.25,2
530,147,1997-02-25,1997-03-06,1997-03-06,10,3550.05,2
36,32,1997-01-31,1997-02-02,1997-01-31,3,772.8,3
41,26,1997-02-02,1997-02-03,1997-02-03,2,627.9,3
46,15,1997-02-05,1997-02-06,1997-02-05,2,362.25,3
55,33,1997-02-10,1997-02-13,1997-02-11,4,796.9499999999999,3
2,3,1997-02-10,1997-02-10,1997-02-10,1,72.44999999999999,4

Using **Script 8 – Category, Number of Days in Category, and Maximum Area in Category (for Region_ID)**, we can calculate the number of days and the maximum area for each category.

	kff	days	area_km2
0	1	62	42117.60
1	2	30	20648.25
2	3	9	796.95
3	4	1	72.45

Note! If there are several objects existing during the same period, the nested objects may be assigned to each of them because the assignment is based only on temporal overlap.

The correct object can be identified by analyzing its area. The area of category 1 should be the largest.